Order-Optimal Noninteractive One-Bit Mean Estimation

July 23, 2026

1 Theoretical Setup

We study estimation of a real mean from one bit per independent sample. The exponent $k > 1$, the range bound λ, the moment scale σ, the target accuracy ϵ, and the confidence level δ are public. A protocol is fully noninteractive if every query set, including every randomized query set, is fixed before the first response is observed.

1.1 Assumptions

Assumption 1 (Parameter domain)**.** *The exponent $k > 1$ is fixed and known, $\lambda \geq \sigma > 0$ are known, $\delta \in (0, 1/2)$, and $0 < \epsilon \leq c_k\sigma$, where $c_k \in (0, 1)$ depends only on k and is fixed in Theorem 3.1.*

Assumption 2 (Unrestricted central moment class)**.** *The samples have a common law D with mean $\mu = \mathbb{E}_D X \in [-\lambda, \lambda]$ and*

$$\mathbb{E}_D|X - \mu|^k \leq \sigma^k.$$

No support, symmetry, density, likelihood, or additional tail condition is imposed.

Assumption 3 (Independent samples and precommitted seeds)**.** *All samples are mutually independent with common law D. All public random variables used below are mutually independent and independent of the samples. The sample split, the median blocks, and every query-generating random variable are fixed before any response bit is observed.*

1.2 Coded localization

The sample indices are split into disjoint localization and refinement sets I_{loc} and I_{ref}. Localization uses confidence $\eta = \delta/4$ and the deterministic coding construction underlying Lau and Scarlett (2026, Theorem 16).

If $2\lambda \leq 20\sigma$, take $I_{\text{loc}} = \varnothing$, issue no localization query, return $I = [-\lambda, \lambda]$, and set $c = 0$. If $2\lambda > 20\sigma$, put

$$N = \left\lceil \frac{2\lambda}{20\sigma} \right\rceil, \qquad \Delta = \frac{2\lambda}{N}, \qquad a_v = -\lambda + (v-1)\Delta \quad (1 \leq v \leq N+1),$$

and define bins

$$K_v = [a_v, a_{v+1}) \quad (v < N), \qquad K_N = [a_N, a_{N+1}].$$

The clipped cells are

$$C_1 = (-\infty, a_2), \qquad C_v = [a_v, a_{v+1}) \ (1 < v < N), \qquad C_N = [a_N, \infty).$$

Let $b(x)$ be the unique cell index satisfying $x \in C_{b(x)}$. Choose the deterministic balanced codewords $\mathsf{c}_1, \dots, \mathsf{c}_N \in \{0,1\}^{N_{\mathrm{loc}}}$ from the cited construction, where

$$N_{\mathrm{loc}} = \left\lceil 10000 \left(\log N + \log \frac{4}{\delta} \right) \right\rceil .$$

For $r \in I_{\mathrm{loc}}$, the precommitted query and response are

$$\mathcal{B}_r = \bigcup_{v:\, \mathsf{c}_{v,r}=1} C_v, \qquad Y_r = \mathbf{1}\{X_r \in \mathcal{B}_r\}.$$

The decoder forms

$$H_v = \sum_{r \in I_{\mathrm{loc}}} \mathbf{1}\{Y_r \neq \mathsf{c}_{v,r}\}, \qquad \widehat{\imath} = \min \operatorname*{argmin}_{v \in [N]} H_v,$$

returns the union of the bins whose indices are within two of $\widehat{\imath}$, clipped to $[N]$, and lets c be the midpoint of that interval. Thus, in both branches, c is always defined and uses no additional query.

1.3 Precommitted refinement bank

Given k-only design constants $\gamma_k \in (0,1)$ and $b_k \geq 1$, define

$$h_0 = \gamma_k \epsilon, \qquad H_* = b_k \sigma \left(\frac{\sigma}{\epsilon} \right)^{1/(k-1)}, \qquad J = \left\lceil \log_2 \frac{H_*}{h_0} \right\rceil,$$

$$h_j = 2^j h_0 \quad (0 \leq j \leq J), \qquad H = h_J.$$

The sampled levels are $0, \dots, J-1$. Split them into

$$\mathcal{J}_{\mathrm{f}} = \{j : h_j \leq \sigma\}, \qquad \mathcal{J}_{\mathrm{c}} = \{j : h_j > \sigma\}.$$

For a nonempty group $G \in \{\mathcal{J}_{\mathrm{f}}, \mathcal{J}_{\mathrm{c}}\}$, put

$$w_j = \begin{cases} h_j/\sigma, & j \in \mathcal{J}_{\mathrm{f}}, \\ (h_j/\sigma)^{2-k}, & j \in \mathcal{J}_{\mathrm{c}}, \end{cases} \qquad W_G = \sum_{\ell \in G} w_\ell.$$

If $m \in \{1,2\}$ groups are nonempty, set

$$p_j = \frac{1}{m} \frac{w_j}{W_G} \qquad (j \in G).$$

An empty group contributes no indices and no normalizer is evaluated.

Let $\mathcal{S} = \{0, 1/4, 1/2, 3/4\}$. For $a, b \in \mathcal{S}$, define

$$Q_{j,a}(x) = a h_j + h_j \left\lfloor \frac{x - a h_j}{h_j} \right\rfloor, \qquad F_{j,a,b}(x) = Q_{j,a}(x) - Q_{j+1,b}(x).$$

Independently for every $i \in I_{\mathrm{ref}}$, draw before communication

$$L_i \sim (p_0, \dots, p_{J-1}), \qquad A_i, B_i \overset{\mathrm{iid}}{\sim} \mathrm{Unif}(\mathcal{S}), \qquad U_i \sim \mathrm{Unif}[-1,2].$$

The encoder sends the single bit

$$Y_i = \mathbf{1}\{X_i \in \mathcal{A}_i\}, \qquad \mathcal{A}_i = \left\{ x : \frac{F_{L_i, A_i, B_i}(x)}{h_{L_i}} \geq U_i \right\}.$$

Every set $\mathcal{A}_i$ is determined by its public seed and contains no occurrence of c.

1.4 Decoder

For every $0 \le j \le J$, let $a_j(c)$ be the unique element of $\mathcal{S}$ satisfying

$$\left\{ \frac{c}{h_j} - a_j(c) \right\} \in [3/8, 5/8),$$

where braces denote fractional part. Write

$$Q_j^c = Q_{j,a_j(c)}, \qquad D_j^c = Q_j^c - Q_{j+1}^c = F_{j,a_j(c),a_{j+1}(c)}.$$

The decoder computes, without another message,

$$\begin{aligned} Z_i(c) = {} & \frac{16}{p_{L_i}} \mathbf{1}\{(A_i, B_i) = (a_{L_i}(c), a_{L_i+1}(c))\} \, 3h_{L_i} \\ & \times \left[Y_i - \mathbf{1}\left\{ \frac{F_{L_i,A_i,B_i}(c)}{h_{L_i}} \ge U_i \right\} \right]. \end{aligned}$$

Choose integers $s, q \ge 1$, with q odd, before communication, set $N_{\mathrm{ref}} = qs$, and partition I_{ref} into fixed blocks $G_1, \dots, G_q$ of size s. With

$$\overline{Z}_g(c) = \frac{1}{s} \sum_{i \in G_g} Z_i(c),$$

the final estimate is

$$\widehat{\mu} = c + \mathrm{median}\big(\overline{Z}_1(c), \dots, \overline{Z}_q(c)\big).$$

2 Preliminaries

For $k > 1$ and $\lambda \ge \sigma > 0$, define

$$\mathcal{D}(k, \lambda, \sigma) := \Big\{ D : \ \mu(D) = \mathbb{E}_D X \in [-\lambda, \lambda], \ \mathbb{E}_D |X - \mu(D)|^k \le \sigma^k \Big\}.$$

All logarithms below are natural. The target rate is

$$r_k(\lambda, \sigma, \epsilon, \delta) := \log \frac{\lambda}{\sigma} + \begin{cases} \dfrac{\sigma^2}{\epsilon^2} \log \dfrac{1}{\delta}, & k > 2, \\ \dfrac{\sigma^2}{\epsilon^2} \log \dfrac{\sigma}{\epsilon} \log \dfrac{1}{\delta}, & k = 2, \\ \left(\dfrac{\sigma}{\epsilon} \right)^{k/(k-1)} \log \dfrac{1}{\delta}, & 1 < k < 2. \end{cases}$$

We write $\Pr_{D,\,\mathrm{protocol}}$ for probability over all samples and all public randomness under the common law D.

3 Main Theorem

Theorem 3.1 (Exact noninteractive refinement certificate)**.** *Let Assumptions 1–3 hold with*

$$C_k^{\mathrm{rec}} = 2^{k-1}(1 + 50^k), \qquad \overline{C}_k^{\mathrm{tail}} = \frac{11}{3} \left(\frac{8}{3} \right)^{k-1} C_k^{\mathrm{rec}},$$

$$c_k = e^{-1}, \qquad \gamma_k = \frac{1}{8}, \qquad b_k = \max\left\{4, (8\overline{C}_k^{\mathrm{tail}})^{1/(k-1)}\right\}.$$

Let

$$C_k^{\mathrm{act}} = \frac{3(8/3)^k}{1-2^{-k}}, \qquad A_k = 96 C_k^{\mathrm{rec}} C_k^{\mathrm{act}},$$

and define the positive k-only coarse constants

$$C_{k,>}^{\mathrm{c}} = \frac{A_k}{1-2^{2-k}} \quad (k>2),$$

$$C_2^{\mathrm{c}} = A_2\left(\log_2(2b_2) + \frac{1}{\log 2}\right) \quad (k=2),$$

$$C_{k,<}^{\mathrm{c}} = \frac{A_k}{2^{2-k}(1-2^{-(2-k)})} \quad (1<k<2).$$

For the protocol of Section 1, set

$$V_k = 3072\sigma^2 \mathbf{1}\{\mathcal{J}_{\mathrm{f}} \neq \varnothing\} + \mathbf{1}\{\mathcal{J}_{\mathrm{c}} \neq \varnothing\} \begin{cases} C_{k,>}^{\mathrm{c}}\sigma^2, & k>2, \\ C_2^{\mathrm{c}}\sigma^2 \log(\sigma/\epsilon), & k=2, \\ C_{k,<}^{\mathrm{c}}\sigma^k H^{2-k}, & 1<k<2, \end{cases}$$

$$s = \left\lceil \frac{32V_k}{\epsilon^2} \right\rceil, \qquad q = 2\left\lceil 8\log\frac{4}{\delta}\right\rceil + 1, \qquad N_{\mathrm{ref}} = qs.$$

Then the complete query bank is Borel and fixed before the first response. The protocol has deterministic horizon

$$n = N_{\mathrm{loc}} + N_{\mathrm{ref}},$$

uses n independent samples, and receives exactly one bit from each sample. Every use of c, including shift selection, centering, and importance weighting, is decoder-only. Moreover,

$$\sup_{D\in\mathcal{D}(k,\lambda,\sigma)} \Pr_{D,\,\mathrm{protocol}}\{|\widehat{\mu} - \mu(D)| > \epsilon\} \le \frac{\delta}{2} \le \delta.$$

The zero-query localization branch remains exact. At $x = c$, every centered digit, residual, and dither bracket is zero; in particular, if D is a point mass at μ and the generated center equals μ, then $\widehat{\mu} = \mu$ exactly.

Corollary 3.1 (Order-optimal noninteractive one-bit mean estimation)**.** *For every fixed $k>1$, use the k-only choices*

$$c_k = e^{-1}, \qquad \gamma_k = \frac{1}{8}, \qquad b_k = \max\left\{4, (8\overline{C}_k^{\mathrm{tail}})^{1/(k-1)}\right\}$$

from Theorem 3.1, together with its displayed choices of s and q. There is a finite constant $C_k > 0$, depending only on k, such that for every known $\lambda \ge \sigma > 0$, every $0 < \epsilon \le c_k\sigma$, every $\delta \in (0, 1/2)$, and every $D \in \mathcal{D}(k,\lambda,\sigma)$, the resulting fully noninteractive protocol has deterministic fixed horizon, uses exactly one bit from each independent sample, and satisfies

$$n \le C_k r_k(\lambda,\sigma,\epsilon,\delta),$$

$$\sup_{D\in\mathcal{D}(k,\lambda,\sigma)} \Pr_{D,\,\mathrm{protocol}}\{|\widehat{\mu} - \mu(D)| > \epsilon\} \le \delta.$$

The probability statement is unconditional over all samples and protocol randomness and uses absolute error on $\mathbb{R}$.

4 Proof Sketch

The coding scheme of Lau and Scarlett (2026, Theorem 16) first produces an interval of length at most 100σ; taking its midpoint gives an always-defined center c with $|c - \mu| \le 50\sigma$ except on an event of probability at most $\delta/4$. Recentring the k-moment at this actual decoder output costs only a k-dependent constant (Lemma A.4).

At each dyadic scale, one of four shifts puts c a distance at least $3h_j/8$ from the selected cell boundary (Proposition A.4). The encoder queries every randomly selected scale-offset entry without knowing c; the decoder retains the matching entry afterward. Uniform dithering turns its centered bit into an exact selected-digit difference (Proposition A.6), and these differences telescope to $X - c$ up to a bottom quantization residual and a top truncation residual (Proposition A.7).

The cell margin makes a digit identically zero until $|X - c| > 3h_j/8$. Summing this activation constraint pathwise before taking expectations gives a count-free fine-scale bound and a k-moment-compatible coarse bound (Propositions A.8 and A.9). Consequently the conditional variance has the three regimes σ^2, $\sigma^2 \log(\sigma/\epsilon)$, and $\sigma^k H^{2-k}$, with only one logarithm at $k = 2$ (Proposition A.16).

The two residual expectations total at most $\epsilon/4$ (Proposition A.17). A fixed median-of-means allocation estimates the exact telescope mean to accuracy $\epsilon/2$, after which the localization and refinement failure probabilities are integrated under their common joint law (Proposition A.22). Finally, the exact ceilings, confidence factors, and localization cost are absorbed into the three-regime rate in the Rate Specialization Bridge, Proposition A.25.

References

Lau, I. and Scarlett, J. (2026). Order-optimal sequential 1-bit mean estimation in general tail regimes. Version 2. 1, 5, 7

A Proof Details

A.1 Coded One-Bit Localization

Lemma A.1 (Central k-moment implies the localization first moment)**.** *Under Assumptions 1 and 2,*

$$\mathbb{E}_D|X-\mu| \le \sigma.$$

Proof. Let $V = |X-\mu|$. Assumption 2 gives $\mathbb{E}V^k \le \sigma^k < \infty$, and Assumption 1 gives $k > 1$. Hölder's inequality with conjugate exponents k and $k/(k-1)$ gives

$$\mathbb{E}V = \mathbb{E}[V \cdot 1] \le (\mathbb{E}V^k)^{1/k}\left(\mathbb{E}1^{k/(k-1)}\right)^{(k-1)/k} = (\mathbb{E}V^k)^{1/k} \le \sigma.$$

This is the first-moment condition required by the localization theorem. □

Proposition A.1 (Deterministic coded localization interval)**.** *Under Assumptions 1, 2, and 3, and using Lemma A.1, set $\eta = \delta/4$. The localization theorem of Lau and Scarlett (2026, Theorem 16 and its localization appendix) has an exact instantiation in the present precommitted Borel-set query convention which uses one bit from every localization sample and returns an always-defined interval I satisfying*

$$|I| \le 100\sigma, \qquad \Pr\{\mu \in I\} \ge 1-\eta.$$

In the branch $2\lambda \le 20\sigma$, the instantiation uses no sample and returns $[-\lambda,\lambda]$. In the branch $2\lambda > 20\sigma$, it uses exactly

$$\ell = \left\lceil 10000\left(\log N + \log\frac{1}{\eta}\right)\right\rceil, \qquad N = \left\lceil \frac{2\lambda}{20\sigma}\right\rceil,$$

localization samples. A deterministic rule resolves every Hamming-score tie.

Proof. First, $\eta = \delta/4 \in (0,1/2)$ by Assumption 1. The same assumption gives $\lambda \ge \sigma > 0$; Assumption 2 gives $\mu \in [-\lambda,\lambda]$; Lemma A.1 gives $\mathbb{E}|X-\mu| \le \sigma$; and Assumption 3 supplies the iid localization samples. These discharge every hypothesis of the cited localization result.

For completeness, the exact object map is as follows. Put $h = 20\sigma$.

Trivial branch $2\lambda \le h$. Take $I_{\mathrm{loc}} = \varnothing$, make the localization public seed a constant, issue no query, and set

$$I = [-\lambda,\lambda].$$

Then I is always defined, $\mu \in I$ deterministically, and

$$|I| = 2\lambda \le h = 20\sigma \le 100\sigma.$$

This includes the boundary $2\lambda = h$, as well as $\lambda = \sigma$.

Nontrivial branch $2\lambda > h$. Define

$$N = \left\lceil \frac{2\lambda}{h}\right\rceil \ge 2, \qquad \Delta = \frac{2\lambda}{N}, \qquad a_v = -\lambda + (v-1)\Delta.$$

Use the localization bins

$$B_v = [a_v, a_{v+1}) \quad (v < N), \qquad B_N = [a_N, a_{N+1}],$$

and the clipped cells

$$C_1 = (-\infty, a_2), \qquad C_v = [a_v, a_{v+1})\ (1 < v < N), \qquad C_N = [a_N, \infty).$$

Thus the clipped-bin map $b : \mathbb{R} \to [N]$ satisfies $b^{-1}(v) = C_v$. Set

$$\ell = \left\lceil 10000 \left(\log N + \log \frac{1}{\eta} \right) \right\rceil .$$

Since $N \geq 2$ and $\ell \geq 10000 \log N$, the checked balanced-codebook lemma supplies deterministic $\mathsf{c}_1, \ldots, \mathsf{c}_N \in \{0,1\}^\ell$. For the t-th localization sample define

$$Q_t(x) = \mathsf{c}_{b(x),t}, \qquad \mathcal{B}_t = Q_t^{-1}(\{1\}) = \bigcup_{v:\, \mathsf{c}_{v,t}=1} C_v.$$

Every C_v is an interval and the union is finite, so $\mathcal{B}_t$ is Borel. Moreover

$$Y_t = Q_t(X_t) = \mathbf{1}\{X_t \in \mathcal{B}_t\}.$$

The codebook, the sets $\mathcal{B}_t$, and the enumeration of I_{loc} depend only on known parameters and are fixed before any sample response. Hence the cited Boolean-query convention and the present measurable-set convention are identical, query by query. Take R_{loc} to be a constant encoding this fixed schedule; its degeneracy satisfies the stipulated independence.

For candidate v, form

$$H_v = \sum_{t=1}^{\ell} \mathbf{1}\{Y_t \neq \mathsf{c}_{v,t}\}, \qquad \widehat{i} = \min \operatorname*{argmin}_{v \in [N]} H_v,$$

and return

$$I = \bigcup_{u=\max\{1,\widehat{i}-2\}}^{\min\{N,\widehat{i}+2\}} B_u.$$

The finite argmin is nonempty, and taking its least index makes both $\widehat{i}$ and I deterministic and always defined for every bit transcript.

The cited appendix proves $h/2 \leq \Delta \leq h$; therefore this union of at most five consecutive bins obeys

$$|I| \leq 5\Delta \leq 5h = 100\sigma.$$

It also proves the containment probability in precisely this convention. In brief, for the true bin $i = b(\mu)$, the safe set $S = \{u : |u-i| \leq 2\}$ has probability mass at least $19/20$ by Markov's inequality and Lemma A.1. A heaviest safe anchor has mass at least $19/100$. The balanced-codeword bounds then give, for every far bin $j \notin S$,

$$\mathbb{E}[H_j - H_{i^*}] > 0.05\ell, \qquad \Pr\{H_j \leq H_{i^*}\} \leq e^{-0.00125\ell}.$$

A union bound yields

$$\Pr\{\widehat{i} \notin S\} \leq Ne^{-0.00125\ell} \leq \eta.$$

The non-strict comparison $H_j \leq H_{i^*}$ already includes every tie between a far minimizer and the safe anchor. Consequently the bound holds for the least-index deterministic tie rule fixed above, rather than requiring an unspecified favorable tie resolution. Whenever $\widehat{i} \in S$, the enlarged interval contains the true bin and hence μ. Thus $\Pr\{\mu \in I\} \geq 1 - \eta$.

Combining the two branches proves the proposition. $\square$

Lemma A.2 (midpoint localization wrapper)**.** *Under Assumptions 1, 2, and 3, and Proposition A.1, define*

$$c = \mathrm{mid}(I) := \frac{\inf I + \sup I}{2}, \qquad \mathcal{E}_{\mathrm{loc}} := \{|c - \mu| \le 50\sigma\}.$$

Then c is an always-defined measurable function of the localization transcript,

$$\Pr(\mathcal{E}_{\mathrm{loc}}) \ge 1 - \frac{\delta}{4},$$

and on $\mathcal{E}_{\mathrm{loc}}$, $|c-\mu| \le 50\sigma$. The midpoint postprocessing uses no additional sample or query.

Proof. Proposition A.1 returns in each branch a nonempty bounded interval I, including the possible half-open interval produced by a consecutive union of localization bins. Hence $\inf I$, $\sup I$, and their midpoint are defined for every transcript. In the nontrivial branch, the decoded index takes values in a finite set and the interval endpoints are deterministic functions of that index; in the trivial branch I is deterministic. Thus c is measurable and always defined.

On the interval-containment event $\{\mu \in I\}$, elementary interval geometry gives

$$|c - \mu| \le \frac{\sup I - \inf I}{2} = \frac{|I|}{2} \le 50\sigma.$$

Therefore $\{\mu \in I\} \subseteq \mathcal{E}_{\mathrm{loc}}$, and Proposition A.1, with $\eta = \delta/4$, yields

$$\Pr(\mathcal{E}_{\mathrm{loc}}) \ge \Pr\{\mu \in I\} \ge 1 - \eta = 1 - \frac{\delta}{4}.$$

In the trivial branch, the stronger deterministic identities $c = 0$ and $|c - \mu| \le \lambda \le 10\sigma$ hold. Defining the present decoder $\mathsf{Dec}_{\mathrm{loc}}$ to perform this midpoint postprocessing does not change a localization query or its sample count. Thus the localization constant may be fixed as the universal value $L_k = 50$. □

Lemma A.3 (explicit localization cost)**.** *Under Assumption 1 and Proposition A.1, the exact localization cost is*

$$N_{\mathrm{loc}} = \begin{cases} 0, & 2\lambda \le 20\sigma, \\ \left\lceil 10000 \left(\log \left\lceil \dfrac{2\lambda}{20\sigma} \right\rceil + \log \dfrac{4}{\delta} \right) \right\rceil, & 2\lambda > 20\sigma, \end{cases}$$

and, uniformly over both branches,

$$N_{\mathrm{loc}} \le 1 + 10000 \log \frac{\lambda}{\sigma} + 30000 \log \frac{1}{\delta} = O\left(1 + \log \frac{\lambda}{\sigma} + \log \frac{1}{\delta}\right).$$

The hidden constant is universal and in particular is independent of $k, \lambda, \sigma, \delta, D$.

Proof. Only the nontrivial branch needs an estimate. Write $\rho = \lambda/\sigma$. Then $\rho > 10$, and

$$N = \left\lceil \frac{\rho}{10} \right\rceil \le \frac{\rho}{10} + 1 \le \rho,$$

so $\log N \le \log \rho$. Also, since $\delta < 1/2$,

$$\log \frac{4}{\delta} = \log 4 + \log \frac{1}{\delta} \le 3 \log \frac{1}{\delta}.$$

Using $\lceil x \rceil \le x + 1$ in the exact appendix count gives

$$N_{\mathrm{loc}} = \ell \le 1 + 10000 \log \frac{\lambda}{\sigma} + 30000 \log \frac{1}{\delta}.$$

The trivial branch has $N_{\mathrm{loc}} = 0$, so the same nonnegative upper bound applies. □

Completion of the localization certificate. Lemma A.1 derives $\mathbb{E}|X-\mu| \le \sigma$ from the unrestricted central k-moment assumption. Together with the parameter and iid conditions, this discharges every hypothesis of the cited localization theorem.

Proposition A.1 then instantiates that theorem and its appendix in the present model: Boolean queries become precommitted Borel membership queries, the localization seed is degenerate, one fresh iid sample yields one bit, the zero-query and coded branches are both covered, and the least-index rule makes every decoder tie deterministic. It gives an always-defined interval with

$$|I| \le 100\sigma, \qquad \Pr\{\mu \in I\} \ge 1-\delta/4.$$

Lemma A.2 supplies the interval-to-midpoint bridge: with $c = \mathrm{mid}(I)$, interval containment implies $|c-\mu| \le 50\sigma$, and hence

$$\Pr(\mathcal{E}_{\mathrm{loc}}) \ge 1-\delta/4.$$

Lemma A.3 gives the exact branchwise count and the uniform additive bound

$$N_{\mathrm{loc}} = O\left(1 + \log\frac{\lambda}{\sigma} + \log\frac{1}{\delta}\right),$$

with a universal hidden constant. These four named results jointly establish the localization center, event, and count, with I retained only for the interval construction. □

A.2 Moment Recentring at the Decoder

Lemma A.4 (actual-center moment recentering)**.** *Under Assumption 2 and Lemma A.2, for every localization transcript whose generated center c lies in $\mathcal{E}_{\mathrm{loc}} = \{|c-\mu| \le 50\sigma\}$,*

$$\mathbb{E}_D|X-c|^k \le 2^{k-1}(1+50^k)\sigma^k.$$

Proof. Fix an arbitrary localization transcript in $\mathcal{E}_{\mathrm{loc}}$. Relative to the distributional expectation $\mathbb{E}_D$, its output c is now a fixed real number. For every $x \in \mathbb{R}$, the scalar triangle inequality and the elementary power inequality $(u+v)^k \le 2^{k-1}(u^k+v^k)$ give

$$\begin{aligned} |x-c|^k &= |(x-\mu)+(\mu-c)|^k \\ &\le \big(|x-\mu|+|\mu-c|\big)^k \\ &\le 2^{k-1}\left(|x-\mu|^k+|\mu-c|^k\right). \end{aligned}$$

Taking expectation with respect to $X \sim D$ is legitimate because the right-hand side is integrable: its first term has finite expectation by Assumption 2, and its second term is a finite constant for the fixed transcript. Hence

$$\mathbb{E}_D|X-c|^k \le 2^{k-1}\left(\mathbb{E}_D|X-\mu|^k+|\mu-c|^k\right).$$

Assumption 2 and the localization bound on $\mathcal{E}_{\mathrm{loc}}$ now yield

$$\begin{aligned} \mathbb{E}_D|X-c|^k &\le 2^{k-1}\left(\sigma^k+(50\sigma)^k\right) \\ &= 2^{k-1}(1+50^k)\sigma^k. \end{aligned}$$

The argument holds for every transcript in the generated event, so the claimed on-event statement is pathwise in the localization output.

For the zero-displacement boundary $c = \mu$, the original expression itself gives the sharper exact specialization

$$\mathbb{E}_D|X - c|^k = \mathbb{E}_D|X - \mu|^k \leq \sigma^k.$$

Thus recentering preserves the primitive moment when no translation is present; the displayed exported constant only covers the full localization radius. $\square$

Completion of the recentered-moment certificate. Lemma A.2 supplies the actual center used by the protocol and proves the derived displacement invariant $|c - \mu| \leq 50\sigma$ on the generated event $\mathcal{E}_{\mathrm{loc}}$. Assumption 2 supplies the only primitive moment input, $\mathbb{E}_D|X - \mu|^k \leq \sigma^k$.

Lemma A.4 composes exactly those two inputs through the direct power inequality and proves, for every localization outcome in $\mathcal{E}_{\mathrm{loc}}$,

$$\mathbb{E}_D|X - c|^k \leq C_k^{\mathrm{rec}}\sigma^k, \qquad C_k^{\mathrm{rec}} := 2^{k-1}(1 + 50^k).$$

The constant is finite for every fixed $k > 1$ and depends only on k. The center in this conclusion is the actual decoder output of Lemma A.2, not an oracle center or a surrogate. This is the stated result and exports the requested recentered moment. $\square$

A.3 Dyadic Scales and Level Probabilities

Lemma A.5 (tail-calibrated dyadic design and ceiling ordering)**.** *Under Assumption 1 and the C_k^{rec} interface of Lemma A.4, define*

$$\overline{C}_k^{\mathrm{tail}} := \frac{11}{3}\left(\frac{8}{3}\right)^{k-1} C_k^{\mathrm{rec}}, \qquad \gamma_k := \frac{1}{8},$$

$$b_k := \max\left\{4, \left(8\overline{C}_k^{\mathrm{tail}}\right)^{1/(k-1)}\right\}, \qquad c_k := e^{-1}.$$

For every $0 < \epsilon \leq c_k\sigma$, let h_0, H_*, J, h_j, H be exactly the setting-defined scales. Then all three design constants are legal, $J \geq 1$, and

$$0 < h_0 < \sigma < \frac{H_*}{2} \leq h_{J-1} = \frac{H}{2} < H_* \leq H < 2H_*.$$

In particular, $0 \in \mathcal{J}_{\mathrm{f}}$, $J - 1 \in \mathcal{J}_{\mathrm{c}}$, and both groups are nonempty. These conclusions remain valid at the largest permitted accuracy $\epsilon = c_k\sigma$.

Proof. Lemma A.4 gives $0 < C_k^{\mathrm{rec}} < \infty$ as a function of fixed k only. Hence $\overline{C}_k^{\mathrm{tail}}$ is positive, finite, and k-only. Because $k - 1 > 0$, the displayed choices satisfy

$$0 < \gamma_k < 1, \qquad b_k \geq 4 \geq 1, \qquad 0 < c_k < 1, \qquad b_k^{k-1} \geq 8\overline{C}_k^{\mathrm{tail}}.$$

Thus they meet every design-domain requirement in the setup.

Put, only for this proof,

$$t := \frac{\sigma}{\epsilon}.$$

Assumption 1 and $c_k = e^{-1}$ imply $t \geq e$, including equality at the largest allowed ϵ. The bottom scale therefore obeys

$$h_0 = \frac{\epsilon}{8} = \frac{\sigma}{8t} \leq \frac{\sigma}{8e} < \sigma.$$

Moreover,

$$\frac{H_*}{h_0} = \frac{b_k \sigma t^{1/(k-1)}}{\epsilon/8} = 8b_k t^{1+1/(k-1)} = 8b_k t^{k/(k-1)} > 1.$$

Consequently $J = \lceil \log_2(H_*/h_0) \rceil$ is a positive integer. Applying the ceiling relation to $x = H_*/h_0$ gives

$$2^{J-1} < \frac{H_*}{h_0} \le 2^J.$$

After multiplying by h_0, and recalling $H = 2^J h_0$, this is

$$\frac{H}{2} = h_{J-1} < H_* \le H < 2H_*.$$

The middle lower bound follows separately from $H \ge H_*$:

$$h_{J-1} = \frac{H}{2} \ge \frac{H_*}{2} = \frac{b_k}{2} \sigma t^{1/(k-1)} \ge 2\sigma e^{1/(k-1)} > \sigma.$$

Together with $h_0 < \sigma$, these displays prove the full ordering and show that the endpoint indices 0 and $J-1$ lie in the claimed groups. Every inequality used $t \ge e$, so equality $\epsilon = c_k \sigma$ causes no endpoint or ceiling failure. □

Proposition A.2 (group partition and empty-group-safe level law)**.** *Let $k > 1$, $\sigma > 0$, $h_0 > 0$, and $J \ge 1$, and set $h_j = 2^j h_0$ for $0 \le j \le J$. With the setting's group, weight, and probability definitions, the sampled indices $\{0, \dots, J-1\}$ are the disjoint union of a fine prefix $\mathcal{J}_{\mathrm{f}}$ and a coarse suffix $\mathcal{J}_{\mathrm{c}}$, with the equality case $h_j = \sigma$ assigned to the fine prefix. Every nonempty group G has $W_G > 0$. If m is the number of nonempty groups and*

$$p_j = \frac{1}{m} \frac{w_j}{W_G} \qquad (j \in G),$$

then

$$\sum_{j \in G} p_j = \frac{1}{m} \quad \textit{for each nonempty } G, \qquad \sum_{j=0}^{J-1} p_j = 1.$$

For inequality statements, an empty group's normalizer may be extended by the empty-sum convention $W_G = 0$; its probability formula is not evaluated. If one group is empty, the other group receives all mass with $m = 1$. Both groups cannot be empty because $J \ge 1$. If, in addition, Assumption 1 holds and the constants and scales are those of Lemma A.5, then both groups are nonempty, so $m = 2$ and each receives mass $1/2$.

Proof. The scales are strictly increasing because $h_{j+1} = 2h_j$. Hence once $h_j > \sigma$, every later sampled scale is also larger than σ, and every earlier scale is smaller than that scale. Thus the indices satisfying $h_j \le \sigma$ form a prefix, while those satisfying $h_j > \sigma$ form the complementary suffix. The weak and strict inequalities make the two sets disjoint and place an exact transition $h_j = \sigma$ in $\mathcal{J}_{\mathrm{f}}$.

Every weight is strictly positive: on the fine group, $w_j = h_j/\sigma > 0$, and on the coarse group, $w_j = (h_j/\sigma)^{2-k} > 0$. A finite sum of at least one positive weight is positive, so $W_G > 0$ for each nonempty group. Consequently the displayed formula for p_j never divides by zero. Direct summation within such a group gives

$$\sum_{j \in G} p_j = \frac{1}{m} \frac{\sum_{j \in G} w_j}{W_G} = \frac{1}{m}.$$

Summing this identity over the m nonempty groups gives total mass one.

If a group is empty, its defining sum contains no terms, so setting that sum to zero is the usual empty-sum extension for bounds. The protocol definition only applies w_j/W_G for $j \in G$, so an empty group creates neither an index nor a denominator. Since $J \geq 1$, the sampled index set is nonempty, and at least one of the two partition sets is nonempty. Thus $m \in \{1, 2\}$, and in the one-group case the preceding mass identity gives mass one to the existing group. Finally, Lemma A.5 proves that index 0 is fine and index $J-1$ is coarse under the theorem choices, so there $m = 2$. $\square$

Lemma A.6 (uniform fine normalizer)**.** *For every positive dyadic scale family covered by Proposition A.2, the fine normalizer satisfies*

$$0 \leq W_\mathrm{f} \leq 2,$$

where $W_\mathrm{f} = 0$ is the empty-sum value if $\mathcal{J}_\mathrm{f} = \varnothing$. If the group is nonempty, then in fact $W_\mathrm{f} < 2$.

Proof. The empty case is immediate. Otherwise let $j_\mathrm{f} := \max \mathcal{J}_\mathrm{f}$. By the prefix property from Proposition A.2, $\mathcal{J}_\mathrm{f} = \{0, \ldots, j_\mathrm{f}\}$. Reindexing backward from its largest scale gives the explicit finite geometric sum

$$\begin{aligned} W_\mathrm{f} = \sum_{j=0}^{j_\mathrm{f}} \frac{h_j}{\sigma} &= \frac{h_{j_\mathrm{f}}}{\sigma} \sum_{r=0}^{j_\mathrm{f}} 2^{-r} \\ &= \frac{h_{j_\mathrm{f}}}{\sigma} \frac{1 - 2^{-(j_\mathrm{f}+1)}}{1 - 1/2} < 2\frac{h_{j_\mathrm{f}}}{\sigma} \leq 2. \end{aligned}$$

The last inequality uses the defining fine condition $h_{j_\mathrm{f}} \leq \sigma$, including equality at the fine/coarse boundary. No bound on the number of fine levels was used. $\square$

Proposition A.3 (three-regime coarse normalizers)**.** *With the setting's empty-group semantics, an empty coarse group has the empty-sum value $W_\mathrm{c} = 0$, and no probability denominator for that group is evaluated. Under Assumption 1, Lemma A.5, and Proposition A.2, put $t = \sigma/\epsilon \geq e$. The theorem scales satisfy*

$$W_\mathrm{c} \leq \begin{cases} \dfrac{1}{1 - 2^{2-k}}, & k > 2, \\ \left(\log_2(2b_k) + \dfrac{1}{\log 2}\right) \log t, & k = 2, \\ \dfrac{1}{1 - 2^{-(2-k)}} \left(\dfrac{H}{2\sigma}\right)^{2-k}, & 1 < k < 2. \end{cases}$$

Here log *is natural, as in the setting. In the last regime this also implies the two structural consequences*

$$W_\mathrm{c} \leq \frac{1}{1 - 2^{-(2-k)}} \left(\frac{H}{\sigma}\right)^{2-k} < \frac{(2b_k)^{2-k}}{1 - 2^{-(2-k)}} t^{(2-k)/(k-1)}.$$

Thus the only nonconstant scale accumulation at $k = 2$ is the explicit $\log(\sigma/\epsilon)$ *factor, while for $1 < k < 2$ the exact structural factor is $(H/\sigma)^{2-k}$.*

Proof. The empty case satisfies every displayed bound because each right-hand side is positive. Suppose henceforth that the coarse group is nonempty. By Proposition A.2, it is a suffix. Write

$$j_\mathrm{c} := \min \mathcal{J}_\mathrm{c}, \qquad n_\mathrm{c} := J - j_\mathrm{c} \geq 1,$$

so its indices are $j_\mathrm{c}, \ldots, J-1$.

If $k > 2$, let $\beta = k - 2 > 0$. Since $h_{j_c}/\sigma > 1$, the first coarse weight satisfies $(h_{j_c}/\sigma)^{-\beta} < 1$. Successive weights have ratio $2^{-\beta} \in (0, 1)$. Therefore the explicit decreasing geometric sum obeys

$$\begin{aligned} W_c &= \left(\frac{h_{j_c}}{\sigma}\right)^{-\beta} \sum_{r=0}^{n_c-1} 2^{-\beta r} \\ &< \sum_{r=0}^{\infty} 2^{-\beta r} = \frac{1}{1-2^{-\beta}} = \frac{1}{1-2^{2-k}}. \end{aligned}$$

This is finite for each fixed $k > 2$ and contains no dependence on J, H, σ, or ϵ.

If $k = 2$, every coarse weight equals one, so $W_c = n_c$ exactly. The largest sampled coarse scale is $h_{J-1} = H/2$, and

$$\frac{H}{2} = h_{J-1} = 2^{n_c-1} h_{j_c} > 2^{n_c-1}\sigma.$$

Thus $n_c < \log_2(H/\sigma)$. At $k = 2$, the ceiling upper bound from Lemma A.5 reads

$$H < 2H_* = 2b_k\sigma t,$$

and hence

$$W_c < \log_2(2b_k t) = \log_2(2b_k) + \frac{\log t}{\log 2}.$$

Since $t \geq e$, $\log t \geq 1$, so the constant first term may be dominated by its explicit multiple of $\log t$:

$$\log_2(2b_k) + \frac{\log t}{\log 2} \leq \left(\log_2(2b_k) + \frac{1}{\log 2}\right) \log t.$$

This verifies the exact $k = 2$ logarithm even when $\epsilon = c_k\sigma$, where $\log t = 1$; no unquantified $+1$ is discarded.

Finally, let $1 < k < 2$ and $\alpha = 2 - k > 0$. Successive coarse weights now have ratio $2^{\alpha} > 1$. Reindexing backward from the largest sampled coarse scale gives

$$\begin{aligned} W_c &= \left(\frac{h_{J-1}}{\sigma}\right)^{\alpha} \sum_{r=0}^{n_c-1} 2^{-\alpha r} \\ &\leq \left(\frac{H}{2\sigma}\right)^{\alpha} \sum_{r=0}^{\infty} 2^{-\alpha r} = \frac{(H/(2\sigma))^{\alpha}}{1-2^{-\alpha}}. \end{aligned}$$

Dropping only the factor $2^{-\alpha} \leq 1$ gives the stated $(H/\sigma)^{\alpha}$ bound. The strict ceiling inequality $H < 2b_k\sigma t^{1/(k-1)}$ then gives

$$\left(\frac{H}{\sigma}\right)^{2-k} < (2b_k)^{2-k} t^{(2-k)/(k-1)},$$

which is the final displayed consequence. All geometric denominators are positive in their stated open regimes. Their possible growth as $k \to 2$ is permitted because k is fixed and hidden constants are only required to be k-dependent. $\square$

Lemma A.7 (Endpoint calibration)**.** *Under Assumption 1, the C_k^{rec} interface of Lemma A.4, and Lemma A.5, the explicit design satisfies*

$$h_0 = \frac{\epsilon}{8}, \qquad \overline{C}_k^{\mathrm{tail}} \frac{\sigma^k}{H^{k-1}} \leq \frac{\epsilon}{8}.$$

Proof. The first equality is the definition $h_0 = \gamma_k \epsilon$ with $\gamma_k = 1/8$. For the top scale, Lemma A.5 gives $H \geq H_*$, while $k-1 > 0$. Therefore

$$H^{k-1} \geq H_*^{k-1} = b_k^{k-1} \sigma^{k-1} \left(\frac{\sigma}{\epsilon}\right) = b_k^{k-1} \frac{\sigma^k}{\epsilon}.$$

It follows that

$$\overline{C}_k^{\text{tail}} \frac{\sigma^k}{H^{k-1}} \leq \frac{\overline{C}_k^{\text{tail}}}{b_k^{k-1}} \epsilon \leq \frac{\epsilon}{8},$$

where the last inequality is exactly the design relation $b_k^{k-1} \geq 8\overline{C}_k^{\text{tail}}$ proved in Lemma A.5. This is only an endpoint-scale certificate; it does not assume or prove any residual-bias statement. □

Completion of the scale and normalizer certificate. Lemma A.4 supplies the explicit finite constant $C_k^{\text{rec}} = 2^{k-1}(1 + 50^k)$. From this bound and Assumption 1, Lemma A.5 chooses

$$\gamma_k = \frac{1}{8}, \quad c_k = e^{-1}, \quad b_k = \max\left\{4, (8\overline{C}_k^{\text{tail}})^{1/(k-1)}\right\}, \quad \overline{C}_k^{\text{tail}} = \frac{11}{3}\left(\frac{8}{3}\right)^{k-1} C_k^{\text{rec}},$$

and proves the exact ceiling relations $H_* \leq H < 2H_*$, $h_0 < \sigma < h_{J-1}$, and $J \geq 1$, including at $\epsilon = c_k \sigma$.

Proposition A.2 then proves the complete group interface. Under the theorem parameters both groups occur and receive mass $1/2$. In every stipulated auxiliary empty-group case, the empty sum is zero, the absent group contributes no indices and creates no denominator, and the remaining group receives mass one. Thus the exact level law is legal for every possible group configuration.

Lemma A.6 and Proposition A.3 give, without hidden level-count terms,

$$W_{\text{f}} \leq 2, \qquad W_{\text{c}} \lesssim_k \begin{cases} 1, & k > 2, \\ \log(\sigma/\epsilon), & k = 2, \\ (H/\sigma)^{2-k}, & 1 < k < 2, \end{cases}$$

where every implicit constant is explicitly displayed in Proposition A.3 and depends only on fixed k. The $k = 2$ proof retains the exact finite level count and uses $\log(\sigma/\epsilon) \geq 1$ to handle the largest allowed epsilon. Lemma A.7 finally exports the two endpoint scale inequalities needed by downstream residual consumers. These named results jointly prove the stated result and export the requested scale/normalizer certificate. □

A.4 Stable Shifted Quantizers

Lemma A.8 (Four-arc stable selector)**.** *Under Assumption 1, fix any $0 \leq j \leq J$. For every $c \in \mathbb{R}$, there exists exactly one $a \in \mathcal{S} = \{0, 1/4, 1/2, 3/4\}$ such that*

$$\left\{\frac{c}{h_j} - a\right\} \in [3/8, 5/8).$$

The resulting function $a_j : \mathbb{R} \to \mathcal{S}$ is Borel. Consequently, for the fixed finite scale family, $c \mapsto (a_0(c), \ldots, a_J(c))$ is Borel as a map into $\mathcal{S}^{J+1}$.

Proof. By Assumption 1 and the setting definition $h_j = 2^j \gamma_k \epsilon$, one has $h_j > 0$. Put

$$t = \left\{\frac{c}{h_j}\right\} \in [0, 1).$$

Subtracting an integer before taking fractional part has no effect, so $\{c/h_j - a\} = \{t - a\}$. A direct reduction modulo one gives the following four phase sets:

a	$\{t \in [0,1) : \{t-a\} \in [3/8, 5/8)\}$
0	$[3/8, 5/8)$
$1/4$	$[5/8, 7/8)$
$1/2$	$[7/8, 1) \cup [0, 1/8)$
$3/4$	$[1/8, 3/8)$.

These four sets are pairwise disjoint and their union is $[0, 1)$. The left-closed, right-open convention assigns each endpoint to exactly one set, so existence and uniqueness hold without an exceptional phase.

The map $c \mapsto \{c/h_j\}$ is Borel because scaling, the floor map, and subtraction are Borel. Each singleton preimage $\{c : a_j(c) = a\}$ is the inverse image of the corresponding Borel phase set in the table. Since $\mathcal{S}$ is finite, a_j is Borel. The coordinatewise product map over the finite index set $0, \ldots, J$ is therefore Borel as well. $\square$

Proposition A.4 (Exact selected-cell margin)**.** *Under Assumption 1 and Lemma A.8, fix $c \in \mathbb{R}$ and $0 \le j \le J$, and let $Q_j^c = Q_{j,a_j(c)}$ be the setting-defined selected quantizer. Then its cell containing c is exactly*

$$[Q_j^c(c), Q_j^c(c) + h_j),$$

and its two boundary distances satisfy

$$c - Q_j^c(c) \in [3h_j/8, 5h_j/8), \qquad Q_j^c(c) + h_j - c \in (3h_j/8, 5h_j/8].$$

In particular,

$$[c - 3h_j/8, c + 3h_j/8] \subseteq [Q_j^c(c), Q_j^c(c) + h_j),$$

so $Q_j^c(x) = Q_j^c(c)$ for every $|x - c| \le 3h_j/8$. At the floor-cell boundaries themselves, the left endpoint is included and the right endpoint is excluded:

$$Q_j^c\big(Q_j^c(c)\big) = Q_j^c(c), \qquad Q_j^c\big(Q_j^c(c) + h_j\big) = Q_j^c(c) + h_j.$$

Proof. Write $a = a_j(c)$,

$$n = \left\lfloor \frac{c}{h_j} - a \right\rfloor, \qquad u = \left\{ \frac{c}{h_j} - a \right\}.$$

By Lemma A.8, $u \in [3/8, 5/8)$. The quantizer definition gives

$$Q_j^c(c) = ah_j + nh_j, \qquad c = Q_j^c(c) + uh_j.$$

Moreover, for any $x \in \mathbb{R}$,

$$Q_j^c(x) = Q_j^c(c) \iff n \le \frac{x}{h_j} - a < n + 1 \iff x \in [Q_j^c(c), Q_j^c(c) + h_j).$$

Thus the left distance is $uh_j \in [3h_j/8, 5h_j/8)$, whereas the right distance is $(1 - u)h_j \in (3h_j/8, 5h_j/8]$. Hence

$$c - 3h_j/8 \ge Q_j^c(c), \qquad c + 3h_j/8 < Q_j^c(c) + h_j.$$

The first comparison is non-strict precisely when $u = 3/8$, and the second is always strict because $u = 5/8$ is excluded. This proves the closed-radius inclusion and quantizer constancy. Substituting the two cell endpoints into the floor definition proves the final two displayed identities and records the left-closed/right-open convention explicitly. ☐

Lemma A.9 (Endpoint and dictionary-grid boundary trace)**.** *Under Assumption 1, Lemma A.8, and Proposition A.4, fix $0 \le j \le J$ and $c \in \mathbb{R}$, and write $t = \{c/h_j\}$. Then:*

1. At every nontrivial endpoint of the four-arc partition, the selector and selected cell are

t	$a_j(c)$	$\{c/h_j - a_j(c)\}$	*selected cell*
$1/8$	$3/4$	$3/8$	$[c - 3h_j/8, c + 5h_j/8)$
$3/8$	0	$3/8$	$[c - 3h_j/8, c + 5h_j/8)$
$5/8$	$1/4$	$3/8$	$[c - 3h_j/8, c + 5h_j/8)$
$7/8$	$1/2$	$3/8$	$[c - 3h_j/8, c + 5h_j/8)$

In each row the shift belonging to the arc immediately to the left would give normalized coordinate $5/8$ and is excluded, while the displayed shift gives $3/8$ and is included. Therefore $c - 3h_j/8$ is the included left cell endpoint, and every x with $|x - c| \le 3h_j/8$, including equality on the left, remains in the selected cell. 2. If c lies on a grid boundary of any candidate quantizer, meaning that for some $a_0 \in \mathcal{S}$ and $m \in \mathbb{Z}$,

$$c = (m + a_0)h_j,$$

then

$$a_j(c) = (a_0 + 1/2) \bmod 1, \qquad \left\{\frac{c}{h_j} - a_j(c)\right\} = 1/2,$$

and the selected cell is exactly $[c - h_j/2, c + h_j/2)$. Thus a boundary for one dictionary grid is the midpoint of the uniquely selected opposite-shift cell. In particular, if $c/h_j \in \mathbb{Z}$ is an unshifted-grid boundary, then $a_j(c) = 1/2$. 3. At the circular seam $t = 0$ (equivalently $t = 1$), the two pieces $[7/8, 1)$ and $[0, 1/8)$ both belong to the single shift $a = 1/2$; the seam itself is assigned uniquely to $a_j(c) = 1/2$, with normalized coordinate $1/2$ and selected cell $[c - h_j/2, c + h_j/2)$.

Proof. The explicit phase partition in Lemma A.8 directly gives the selector in each row of part 1. At each listed phase, subtracting the displayed shift modulo one gives $3/8$; using the shift assigned to the immediately preceding half-open arc gives $5/8$. The latter value is excluded by the right-open stable band. Proposition A.4 then converts the normalized coordinate $3/8$ into the displayed cell and proves inclusion at the left equality.

For part 2, define $a_0^{\oplus} = (a_0 + 1/2) \bmod 1 \in \mathcal{S}$. From $c/h_j = m + a_0$,

$$\left\{\frac{c}{h_j} - a_0^{\oplus}\right\} = 1/2 \in [3/8, 5/8).$$

Uniqueness in Lemma A.8 forces $a_j(c) = a_0^{\oplus}$. The exact cell formula follows from Proposition A.4 with normalized coordinate $1/2$. Taking $a_0 = 0$ proves the unshifted case.

Finally, the $a = 1/2$ row of the four-arc partition is exactly $[7/8, 1) \cup [0, 1/8)$, and $\{0 - 1/2\} = 1/2$. This proves the seam statement and exhausts the remaining endpoint of the phase circle. ☐

Completion of the stable-selector certificate. Lemma A.8 proves, for every real c and every setting scale $j = 0, \ldots, J$, that the four half-open quarter arcs give exactly one shift $a_j(c)$, and that each coordinate selector and the full finite selector vector are Borel. Proposition A.4 converts the selected fractional coordinate into the exact floor-cell certificate

$$c - Q_j^c(c) \in [3h_j/8, 5h_j/8), \qquad Q_j^c(c) + h_j - c \in (3h_j/8, 5h_j/8],$$

and hence gives the exact two-sided $3h_j/8$ margin, with the full closed interval $[c - 3h_j/8, c + 3h_j/8]$ retained in the selected half-open cell. Lemma A.9 checks every partition endpoint, the circular seam, every boundary of each of the four candidate grids, the unshifted-grid special case, and the left-equality/right-strict behavior. Together these results prove the stable-selector certificate deterministically, without a probabilistic event or rate condition. ☐

A.5 Precommitted Dithered Queries

Lemma A.10 (Exact floor residuals and bounded dyadic digits)**.** *For every setting scale $h_j > 0$, every $0 \le j < J$, every $a, b \in \mathcal{S}$, and every $x \in \mathbb{R}$,*

$$0 \le x - Q_{j,a}(x) < h_j, \qquad 0 \le x - Q_{j+1,b}(x) < h_{j+1} = 2h_j,$$

and consequently

$$-h_j < F_{j,a,b}(x) < 2h_j.$$

The two digit-range endpoints are never attained, including when x is a boundary of either shifted grid.

Proof. For a width $s > 0$ and shift $\alpha \in \mathbb{R}$, put

$$n = \left\lfloor \frac{x - \alpha s}{s} \right\rfloor.$$

The defining floor inequalities give

$$n \le \frac{x - \alpha s}{s} < n + 1,$$

so

$$(\alpha + n)s \le x < (\alpha + n + 1)s.$$

Thus the shifted floor quantizer $(\alpha + n)s$ is the included left endpoint of the unique half-open cell containing x, and

$$0 \le x - (\alpha + n)s < s.$$

At an included left endpoint $x = (\alpha + n)s$, the residual is exactly zero. The point $(\alpha + n + 1)s$ is excluded from that cell and included as the left endpoint of the next cell, where the floor index is $n + 1$ and the residual again equals zero. Hence the residual can never equal s; this explicitly covers positive, zero, and negative cell indices.

Apply this calculation first with $(s, \alpha) = (h_j, a)$ and then with $(s, \alpha) = (h_{j+1}, b) = (2h_j, b)$. Writing the two residuals only within this proof as

$$r_j = x - Q_{j,a}(x) \in [0, h_j), \qquad r_{j+1} = x - Q_{j+1,b}(x) \in [0, 2h_j),$$

one obtains

$$F_{j,a,b}(x) = Q_{j,a}(x) - Q_{j+1,b}(x) = r_{j+1} - r_j.$$

Because $r_{j+1} \ge 0$ and $r_j < h_j$, the difference is strictly greater than $-h_j$. Because $r_{j+1} < 2h_j$ and $r_j \ge 0$, it is strictly less than $2h_j$. Equality at $-h_j$ would require the impossible value $r_j = h_j$, and equality at $2h_j$ would require the impossible value $r_{j+1} = 2h_j$. □

Proposition A.5 (Borel and precommitted refinement queries)**.** *Under Assumption 3, Lemma A.8, Proposition A.4, and Lemma A.10, for every realized refinement seed $(L_i, A_i, B_i, U_i) = (j, a, b, u)$, the setting-defined query*

$$\mathcal{A}_i = \left\{ x \in \mathbb{R} : \frac{F_{j,a,b}(x)}{h_j} \ge u \right\}$$

is Borel. The membership map is jointly Borel in (x, j, a, b, u) on $\mathbb{R} \times \{0, \dots, J-1\} \times \mathcal{S}^2 \times [-1, 2]$, and $\mathcal{A}_i$ is fixed before any response bit is observed. Moreover, Lemma A.8 makes the decoder-side selected pair and centering indicator Borel in c, but neither c nor any response bit changes $\mathcal{A}_i$.

Proof. For fixed j, a, b, the two quantizers are constant on the half-open cells

$$C_n = [(a+n)h_j, (a+n+1)h_j), \qquad K_m = [(b+m)h_{j+1}, (b+m+1)h_{j+1}),$$

where $n, m \in \mathbb{Z}$. On $C_n \cap K_m$,

$$F_{j,a,b}(x) = (a+n)h_j - (b+m)h_{j+1}.$$

Consequently,

$$\mathcal{A}_i = \bigcup_{\substack{n,m\in\mathbb{Z}: \\ ((a+n)h_j - (b+m)h_{j+1})/h_j \ge u}} (C_n \cap K_m).$$

Each intersection is empty or a half-open interval, and the union is countable, proving that $\mathcal{A}_i$ is Borel. Equivalently, the floor map is Borel, so $F_{j,a,b}$ is Borel and $\mathcal{A}_i$ is its inverse image of the closed ray $[u, \infty)$ after division by $h_j > 0$.

The map $(x, u) \mapsto F_{j,a,b}(x)/h_j - u$ is Borel. Since the level and offset spaces are finite, adjoining (j, a, b) proves joint Borel measurability of the membership indicator. Lemma A.10 also resolves the dither-support endpoints: if $u = -1$, then $F_{j,a,b}(x)/h_j > -1$ for every x, so $\mathcal{A}_i = \mathbb{R}$; if $u = 2$, then $F_{j,a,b}(x)/h_j < 2$, so $\mathcal{A}_i = \varnothing$. Thus neither endpoint creates a range-overflow or measurability exception.

By Assumption 3, L_i, A_i, B_i, U_i are all generated before any response bit. The displayed set depends only on those seeds and the known, deterministic scale family, and contains no occurrence of the decoded center c. It is therefore a precommitted measurable query.

For completeness on the decoder side, Lemma A.8 gives Borel maps $c \mapsto a_j(c)$ and $c \mapsto a_{j+1}(c)$. Because their ranges are finite and every $F_{j,a,b}$ is Borel, both

$$c \longmapsto F_{j,a_j(c),a_{j+1}(c)}(c) = D_j^c(c)$$

and the corresponding threshold indicator are Borel. Proposition A.4 additionally places c at positive distance from the boundaries of both selected cells, so the selected floor values have no spatial boundary ambiguity. These decoder operations only select and center an already observed, precommitted bit; they do not alter its query set. □

Proposition A.6 (Exact bounded-uniform-dither moments)**.** *Under Assumption 3, Lemma A.8, and Lemma A.10, fix $0 \le j < J$, $a, b \in \mathcal{S}$, and $x, c \in \mathbb{R}$. If $U \sim \mathrm{Unif}[-1, 2]$, then*

$$\mathbb{E}_U\left[3h_j\left(\mathbf{1}\left\{\frac{F_{j,a,b}(x)}{h_j}\geq U\right\}-\mathbf{1}\left\{\frac{F_{j,a,b}(c)}{h_j}\geq U\right\}\right)\right]=F_{j,a,b}(x)-F_{j,a,b}(c),$$

and

$$\mathbb{E}_U\left[\left(3h_j\left(\mathbf{1}\left\{\frac{F_{j,a,b}(x)}{h_j}\geq U\right\}-\mathbf{1}\left\{\frac{F_{j,a,b}(c)}{h_j}\geq U\right\}\right)\right)^2\right]=3h_j\left|F_{j,a,b}(x)-F_{j,a,b}(c)\right|.$$

These identities remain valid in the actual refinement protocol after conditioning on

$$(c,X_i,L_i,A_i,B_i).$$

This tuple is independent of U_i, and the conditioning occurs before the response bit Y_i is revealed. In particular, for the Borel selected pair $(a_j(c),a_{j+1}(c))$ supplied by Lemma A.8,

$$\mathbb{E}_U\left[3h_j\left(\mathbf{1}\left\{\frac{D_j^c(x)}{h_j}\geq U\right\}-\mathbf{1}\left\{\frac{D_j^c(c)}{h_j}\geq U\right\}\right)\right]=D_j^c(x)-D_j^c(c),$$

$$\mathbb{E}_U\left[\left(3h_j\left(\mathbf{1}\left\{\frac{D_j^c(x)}{h_j}\geq U\right\}-\mathbf{1}\left\{\frac{D_j^c(c)}{h_j}\geq U\right\}\right)\right)^2\right]=3h_j\left|D_j^c(x)-D_j^c(c)\right|.$$

At $x=c$, the centered indicator difference and both displayed quantities are zero for every U, not merely almost surely.

Proof. Use the proof-local abbreviations

$$f_x=\frac{F_{j,a,b}(x)}{h_j},\qquad f_c=\frac{F_{j,a,b}(c)}{h_j}.$$

Lemma A.10 gives $f_x,f_c\in(-1,2)$. For any $t\in(-1,2)$, the threshold convention in the query yields

$$\mathbb{E}_U\mathbf{1}\{t\geq U\}=\Pr\{U\leq t\}=\frac{t+1}{3}.$$

Subtracting this equality at $t=f_x$ and $t=f_c$, then multiplying by $3h_j$, proves the first identity. For the square, define only inside this proof

$$\Delta_U=\mathbf{1}\{f_x\geq U\}-\mathbf{1}\{f_c\geq U\}.$$

If $f_x>f_c$, then $\Delta_U=1$ exactly on $(f_c,f_x]$ and is zero elsewhere. If $f_c>f_x$, then $\Delta_U=-1$ exactly on $(f_x,f_c]$ and is zero elsewhere. If $f_x=f_c$, it is identically zero. Hence in all cases

$$\Delta_U^2=\mathbf{1}\{\min(f_x,f_c)<U\leq\max(f_x,f_c)\}.$$

The interval lies inside $[-1,2]$ and has length $|f_x-f_c|$. Since the uniform density is $1/3$,

$$\mathbb{E}_U\Delta_U^2=\frac{|f_x-f_c|}{3}.$$

Multiplication by $9h_j^2$ gives

$$\mathbb{E}_U(3h_j\Delta_U)^2=3h_j^2|f_x-f_c|=3h_j|F_{j,a,b}(x)-F_{j,a,b}(c)|,$$

which is the exact square identity. The half-open interval above resolves threshold equality: at the smaller threshold both indicators equal one, whereas at the larger threshold only the larger-threshold indicator equals one. Such ties are U-null, but the pointwise convention is nevertheless consistent. At the support endpoints $U = -1$ and $U = 2$, both indicator differences vanish because $f_x, f_c \in (-1, 2)$; including those endpoints in the dither support therefore leaves both identities unchanged.

Assumption 3 makes each U_i independent of the localization output c, the refinement sample X_i, and (L_i, A_i, B_i). Therefore, conditional on any realized values of those non-dither quantities, U_i retains the same uniform law and the two calculations apply verbatim. Finally, choosing $a = a_j(c)$ and $b = a_{j+1}(c)$, whose existence and Borel measurability follow from Lemma A.8, turns $F_{j,a,b}$ into the exact setting-defined selected digit D_j^c. When $x = c$, the two threshold indicators are identical for every U, proving the exact zero-displacement specialization. □

Completion of the bounded-dither certificate. Lemma A.10 derives from the half-open floor convention the strict digit range

$$-h_j < F_{j,a,b}(x) < 2h_j$$

for every bank entry and every real input, and it explicitly excludes both apparent endpoint values even at either grid's boundaries. Proposition A.5 converts the same cell description into a Borel threshold set for every realized seed, proves joint membership measurability, and uses Assumption 3 to establish that the query is fixed before all responses. Lemma A.8 supplies the Borel decoder-selected pair, and Proposition A.4 confirms that decoder centering at c has no selected-grid boundary ambiguity, while leaving the precommitted query unchanged.

Proposition A.6 then integrates the exact uniform threshold rule. Its first calculation gives the signed digit difference, and its half-open interval calculation gives the exact square

$$3h_j|F_{j,a,b}(x) - F_{j,a,b}(c)|$$

without an inequality or remainder. Instantiating the Borel pair from Lemma A.8 gives the identical interface for $D_j^c(x) - D_j^c(c)$, and $x = c$ gives pathwise zero for every dither value. These three named results jointly prove the stated result and export only the bounded digit range, Borel precommitted-query certificate, and exact dither mean/square identities needed by the downstream conditional-moment step. □

A.6 Telescope and Residual Identities

Lemma A.11 (Finite selected-quantizer telescope)**.** *Under the setting definitions and the stable-selector certificate from Lemma A.8, Proposition A.4, and Lemma A.9, for every $c, x \in \mathbb{R}$,*

$$T_c(x) = \big(Q_0^c(x) - Q_0^c(c)\big) - \big(Q_J^c(x) - Q_J^c(c)\big).$$

Proof. Lemmas A.8 and A.9 make every Q_j^c appearing below well defined. By the setting definition $D_j^c = Q_j^c - Q_{j+1}^c$, each summand in $T_c(x)$ is

$$\begin{aligned} D_j^c(x) - D_j^c(c) &= \big(Q_j^c(x) - Q_{j+1}^c(x)\big) - \big(Q_j^c(c) - Q_{j+1}^c(c)\big) \\ &= \big(Q_j^c(x) - Q_j^c(c)\big) - \big(Q_{j+1}^c(x) - Q_{j+1}^c(c)\big). \end{aligned}$$

Therefore, for $J \geq 1$, the finite sum expands as

$$\begin{aligned}T_c(x) &= \sum_{j=0}^{J-1}\bigl(Q_j^c(x) - Q_j^c(c)\bigr) - \sum_{j=0}^{J-1}\bigl(Q_{j+1}^c(x) - Q_{j+1}^c(c)\bigr)\\ &= \bigl(Q_0^c(x) - Q_0^c(c)\bigr) - \bigl(Q_J^c(x) - Q_J^c(c)\bigr),\end{aligned}$$

because every term with index $1, \dots, J-1$ occurs once with each sign. If the formally degenerate case $J = 0$ is allowed, the defining sum is empty and both endpoint terms on the right are the same, so the identity remains valid. In particular, the stated identity includes the boundary case $J = 1$, where there are no intermediate terms and the single summand already equals the two endpoints. No limiting argument or infinite rearrangement is involved. □

Lemma A.12 (Floor-quantizer remainder range)**.** *Under the setting definitions and the stable-selector certificate from Lemma A.8, Proposition A.4, and Lemma A.9, fix $c, y \in \mathbb{R}$ and $0 \le j \le J$. Then*

$$0 \le y - Q_j^c(y) < h_j.$$

Consequently, for every $c, x \in \mathbb{R}$,

$$-h_0 < R_0^c(x) < h_0, \qquad \textit{and hence} \qquad |R_0^c(x)| < h_0.$$

Proof. Put $a = a_j(c)$, which is uniquely defined by Lemma A.8, and set

$$v = \frac{y - ah_j}{h_j}.$$

The defining property of the floor function is

$$\lfloor v \rfloor \le v < \lfloor v \rfloor + 1.$$

Because $h_j > 0$, multiplication by h_j preserves both inequality directions. Using

$$Q_j^c(y) = ah_j + h_j \lfloor v \rfloor$$

gives

$$0 \le y - Q_j^c(y) < h_j.$$

This calculation is valid for every real v, so it includes negative quantizer cells and points exactly on a shifted grid, where the remainder is 0.

At scale zero, write

$$r_x = x - Q_0^c(x), \qquad r_c = c - Q_0^c(c).$$

The first part gives $r_x, r_c \in [0, h_0)$. Hence

$$-h_0 < -r_c \le r_x - r_c \le r_x < h_0.$$

Since $R_0^c(x) = r_x - r_c$, the claimed strict two-sided and absolute-value bounds follow. The strictness is preserved even when one remainder is zero; no cell-margin estimate is needed. □

Proposition A.7 (Residual interface)**.** *Under the setting definitions, Lemma A.8, Proposition A.4, and Lemma A.9 give the stable-selector certificate. Together with Lemmas A.11 and A.12, it yields, for every $c, x \in \mathbb{R}$,*

$$x - c = T_c(x) + R_0^c(x) + R_H^c(x), \qquad |R_0^c(x)| < h_0.$$

Both residuals in the displayed equality are retained as separate signed terms. Moreover, if X is any integrable real random variable, then all terms below are integrable and

$$\mathbb{E}(X - c) = \mathbb{E}T_c(X) + \mathbb{E}R_0^c(X) + \mathbb{E}R_H^c(X).$$

Thus, whenever $\mu = \mathbb{E}X$, the exact telescope-to-mean discrepancy is

$$(\mu - c) - \mathbb{E}T_c(X) = \mathbb{E}R_0^c(X) + \mathbb{E}R_H^c(X),$$

with neither bias term discarded, merged into the other, or bounded in this step.

Proof. Add and subtract the scale-zero selected quantizer values:

$$\begin{aligned} x - c &= \big[(x - Q_0^c(x)) - (c - Q_0^c(c))\big] + \big[Q_0^c(x) - Q_0^c(c)\big] \\ &= R_0^c(x) + \big[Q_0^c(x) - Q_0^c(c)\big]. \end{aligned}$$

Lemma A.11 rearranges, without approximation, to

$$Q_0^c(x) - Q_0^c(c) = T_c(x) + \big[Q_J^c(x) - Q_J^c(c)\big] = T_c(x) + R_H^c(x).$$

Substitution proves

$$x - c = T_c(x) + R_0^c(x) + R_H^c(x).$$

The strict fine-residual bound is exactly Lemma A.12. Notice that the algebra does not replace $R_H^c(x)$ by zero and does not combine it with $R_0^c(x)$.

For the expectation export, suppose X is integrable. Lemma A.12 gives

$$Q_j^c(X) = X - \big(X - Q_j^c(X)\big), \qquad 0 \le X - Q_j^c(X) < h_j$$

for $j = 0, J$. Thus $Q_0^c(X)$ is integrable, and so is $Q_J^c(X)$. The endpoint representation in Lemma A.11 then makes $T_c(X)$ integrable. The top residual is an integrable difference involving $Q_J^c(X)$, while the fine residual is bounded in absolute value by h_0. Taking expectations in the already proved pointwise equality is therefore legitimate and yields the displayed formulas. This expectation passage introduces no moment estimate, inactivity statement, probability event, or residual bound beyond the strict floor-remainder bound proved here. □

Lemma A.13 (Exact zero-displacement specialization)**.** *Under the setting definitions and the stable-selector certificate from Lemma A.8, Proposition A.4, and Lemma A.9, for every $c \in \mathbb{R}$ and every $0 \le j < J$,*

$$D_j^c(c) - D_j^c(c) = 0.$$

Consequently,

$$T_c(c) = 0, \qquad R_0^c(c) = 0, \qquad R_H^c(c) = 0,$$

and the exact displacement decomposition at $x = c$ is

$$c - c = T_c(c) + R_0^c(c) + R_H^c(c), \qquad \textit{i.e.} \qquad 0 = 0 + 0 + 0.$$

Proof. Each centered digit summand is the difference of the same real number from itself, so it is zero. The finite sum defining $T_c(c)$ is therefore zero, including the empty-sum case. Directly from the two residual definitions,

$$R_0^c(c) = \big(c - Q_0^c(c)\big) - \big(c - Q_0^c(c)\big) = 0,$$

and

$$R_H^c(c) = Q_J^c(c) - Q_J^c(c) = 0.$$

Substitution gives the exact displayed specialization. This is an identity, not a bound of order h_0 or H. $\square$

Completion of the telescope and residual certificate. Lemma A.11 proves the exact endpoint formula

$$T_c(x) = \big(Q_0^c(x) - Q_0^c(c)\big) - \big(Q_J^c(x) - Q_J^c(c)\big)$$

by a finite signed cancellation in which every intermediate selected quantizer appears once with each sign. Lemma A.12 proves directly from the floor convention that the two scale-zero remainders lie in $[0, h_0)$, and hence that their difference satisfies the required strict bound $|R_0^c(x)| < h_0$. Proposition A.7 combines these two exact statements to obtain

$$x - c = T_c(x) + R_0^c(x) + R_H^c(x),$$

retaining the fine residual and the top residual as distinct signed terms. Its expectation-level export likewise exposes the only two algebraic telescope biases separately as $\mathbb{E}R_0^c(X)$ and $\mathbb{E}R_H^c(X)$, conditional only on integrability and without importing any future digit-moment, inactivity, tail-support, or bias-control claim. Finally, Lemma A.13 proves that every centered digit summand and both residuals vanish exactly at $x = c$, so the required baseline is $0 = 0 + 0 + 0$, not an asymptotic or bounded-remainder surrogate. These named results jointly establish the stated result and export the requested telescope/residual interface. $\square$

A.7 Digit Activation and Pathwise Budgets

Throughout this subsection, write

$$\Delta_j^c(x) := D_j^c(x) - D_j^c(c), \qquad 0 \le j < J,$$

and, whenever c, x are fixed, put $R = |x - c|$. We also use the k-only constant

$$C_k^{\mathrm{act}} := \frac{3(8/3)^k}{1 - 2^{-k}},$$

whose adequacy is proved in Proposition A.9.

Lemma A.14 (Stable two-scale digit inactivity and top-cell support)**.** *Under Proposition A.4, Lemma A.9, and the exact top-residual interface in Proposition A.7, fix $c, x \in \mathbb{R}$ and put $R = |x - c|$. Then, for every $0 \le j < J$,*

$$R \le \frac{3h_j}{8} \quad \Longrightarrow \quad \Delta_j^c(x) = 0.$$

Equivalently,

$$\Delta_j^c(x) = \Delta_j^c(x)\mathbf{1}\left\{R > \frac{3h_j}{8}\right\}.$$

At the top scale,

$$R \le \frac{3H}{8} \quad \Longrightarrow \quad R_H^c(x) = 0, \qquad R_H^c(x) = R_H^c(x)\mathbf{1}\left\{R > \frac{3H}{8}\right\}.$$

These implications include equality for both displacement signs and remain valid when c or x is a shifted-grid boundary.

Proof. Fix $0 \le j < J$ and suppose $R \le 3h_j/8$. Proposition A.4 at scale j gives

$$Q_j^c(x) = Q_j^c(c).$$

The digit also uses scale $j+1$. Since $h_{j+1} = 2h_j$,

$$R \le \frac{3h_j}{8} < \frac{3h_{j+1}}{8} = \frac{3h_j}{4},$$

so the same proposition at scale $j+1$ gives $Q_{j+1}^c(x) = Q_{j+1}^c(c)$. Therefore

$$\begin{aligned}\Delta_j^c(x) &= \big(Q_j^c(x) - Q_{j+1}^c(x)\big) - \big(Q_j^c(c) - Q_{j+1}^c(c)\big)\\ &= 0.\end{aligned}$$

The support-indicator identity follows by splitting into $R \le 3h_j/8$ and $R > 3h_j/8$.

At scale J, $h_J = H$. If $R \le 3H/8$, Proposition A.4 gives $Q_J^c(x) = Q_J^c(c)$. Proposition A.7 then gives $R_H^c(x) = 0$, and the top support-indicator identity follows.

At $R = 3h_j/8$, the scale-j cell contains both endpoints of the closed stable radius: a possible equality is at its included left boundary, while the positive endpoint is strictly below the excluded right boundary. The same displacement is strictly inside the scale-$j+1$ radius $3h_j/4$. The identical left-included/right-strict argument handles $R = 3H/8$. Lemma A.9 shows that a center on any candidate grid is the midpoint of the selected opposite-shift cell, including for negative grid indices. At $x = c$, $R = 0$, and every conclusion is exactly zero, consistently with Lemma A.13. □

Lemma A.15 (Strict centered-digit envelope and active-scale cutoff)**.** *Under Lemmas A.10 and A.14, fix $c, x \in \mathbb{R}$, put $R = |x - c|$, and let $0 \le j < J$. Then*

$$-3h_j < \Delta_j^c(x) < 3h_j, \qquad |\Delta_j^c(x)| < 3h_j.$$

Moreover,

$$\Delta_j^c(x) \ne 0 \quad \Longrightarrow \quad R > \frac{3h_j}{8}, \qquad R > \frac{3h_j}{8} \quad \Longleftrightarrow \quad h_j < \frac{8R}{3},$$

and

$$|\Delta_j^c(x)| \le 3h_j\mathbf{1}\left\{R > \frac{3h_j}{8}\right\} = 3h_j\mathbf{1}\left\{h_j < \frac{8R}{3}\right\}.$$

In particular, a level with $R = 3h_j/8$ is inactive.

Proof. Because $D_j^c = F_{j,a_j(c),a_{j+1}(c)}$, Lemma A.10 gives

$$-h_j < D_j^c(y) < 2h_j \qquad (y = x, c).$$

Thus

$$D_j^c(x) - D_j^c(c) > -h_j - 2h_j = -3h_j, \qquad D_j^c(x) - D_j^c(c) < 2h_j - (-h_j) = 3h_j.$$

This proves the strict magnitude. It remains true at any grid boundary because the range proof resets the appropriate half-open floor remainder to zero and excludes both digit endpoints.

The contrapositive of Lemma A.14 gives $\Delta_j^c(x) \neq 0 \Rightarrow R > 3h_j/8$. Since $h_j > 0$, the latter is equivalent to $h_j < 8R/3$. When $R \leq 3h_j/8$, the envelope has both sides zero; when $R > 3h_j/8$, strict magnitude implies the asserted non-strict envelope. No converse is claimed: after leaving the stable radius, the centered digit may still vanish by cell constancy or cancellation. □

Lemma A.16 (Finite dyadic cutoff summation)**.** *Under $h_j = 2^j h_0$ with $h_0 > 0$, let $\alpha > 0$ and let $\mathcal{A}$ be a finite subset of nonnegative scale indices. If $\mathcal{A} \neq \varnothing$ and $M = \max \mathcal{A}$, then*

$$\sum_{j\in\mathcal{A}} h_j^\alpha \leq \sum_{j=0}^{M} h_j^\alpha = h_M^\alpha \frac{1-2^{-\alpha(M+1)}}{1-2^{-\alpha}} < \frac{h_M^\alpha}{1-2^{-\alpha}}.$$

Consequently, if $h_j < T$ for every $j \in \mathcal{A}$, where $T > 0$, then

$$\sum_{j\in\mathcal{A}} h_j^\alpha < \frac{T^\alpha}{1-2^{-\alpha}}.$$

The same conclusion holds if $h_j \leq T$ for every $j \in \mathcal{A}$. If $\mathcal{A} = \varnothing$, its sum is 0.

Proof. For nonempty $\mathcal{A}$, positivity gives

$$\sum_{j\in\mathcal{A}} h_j^\alpha \leq \sum_{j=0}^{M} h_j^\alpha.$$

Since $h_j = 2^{j-M} h_M$, the change of variables $r = M - j$ gives

$$\sum_{j=0}^{M} h_j^\alpha = h_M^\alpha \sum_{r=0}^{M} 2^{-\alpha r} = h_M^\alpha \frac{1-2^{-\alpha(M+1)}}{1-2^{-\alpha}}.$$

The denominator is positive and the numerator is strictly less than one. If $h_M < T$, substitution gives the strict threshold bound. If only $h_M \leq T$, the already strict finite-geometric-factor inequality still gives a strict bound by $T^\alpha/(1-2^{-\alpha})$. The empty case is the empty-sum convention. The estimate depends on the largest scale, not on $|\mathcal{A}|$. □

Proposition A.8 (Count-free fine activation ledger)**.** *Under Lemma A.15, Lemma A.16, and*

$$\mathcal{J}_{\mathrm{f}} = \{0 \leq j < J : h_j \leq \sigma\},$$

for every $c, x \in \mathbb{R}$,

$$\sum_{j\in\mathcal{J}_{\mathrm{f}}} \left|D_j^c(x) - D_j^c(c)\right| \leq 16 \min\{|x-c|, \sigma\}.$$

This includes an empty fine group, $h_j = \sigma$, $x = c$, and any finite number of fine levels, with a constant independent of J.

Proof. Fix c, x, put $R = |x - c|$, and define

$$\mathcal{A}_{\mathrm{f}}(R) := \left\{ j \in \mathcal{J}_{\mathrm{f}} : h_j < \frac{8R}{3} \right\}.$$

Lemma A.15 gives

$$\sum_{j \in \mathcal{J}_{\mathrm{f}}} |\Delta_j^c(x)| \le 3 \sum_{j \in \mathcal{A}_{\mathrm{f}}(R)} h_j. \tag{7.1}$$

If $R = 0$, the cutoff set and both sides of the target inequality are zero. Suppose $0 < R \le \sigma$. Lemma A.16, with $\alpha = 1$ and $T = 8R/3$, gives

$$\sum_{j \in \mathcal{A}_{\mathrm{f}}(R)} h_j < \frac{8R/3}{1 - 2^{-1}} = \frac{16R}{3}$$

when the cutoff set is nonempty; the required non-strict bound is immediate if it is empty. Hence (7.1) gives

$$\sum_{j \in \mathcal{J}_{\mathrm{f}}} |\Delta_j^c(x)| \le 16R = 16 \min\{R, \sigma\}.$$

Now suppose $R > \sigma$. If $\mathcal{J}_{\mathrm{f}}$ is empty, the claim is immediate. Otherwise $h_j \le \sigma$ for every fine level, and Lemma A.16 with $\alpha = 1$ gives

$$\sum_{j \in \mathcal{J}_{\mathrm{f}}} h_j < 2\sigma.$$

Enlarging the cutoff sum in (7.1) to all fine levels yields

$$\sum_{j \in \mathcal{J}_{\mathrm{f}}} |\Delta_j^c(x)| < 6\sigma \le 16\sigma = 16 \min\{R, \sigma\}.$$

At $R = \sigma$, the first case applies. The second case explicitly permits the included transition level $h_j = \sigma$. No line depends on the number of fine levels. □

Proposition A.9 (Moment-compatible coarse activation ledger)**.** *Under $k > 1$, Lemma A.15, Lemma A.16, and*

$$\mathcal{J}_{\mathrm{c}} = \{0 \le j < J : h_j > \sigma\},$$

for every $c, x \in \mathbb{R}$,

$$\sum_{j \in \mathcal{J}_{\mathrm{c}}} h_j^{k-1} \left| D_j^c(x) - D_j^c(c) \right| \le C_k^{\mathrm{act}} |x - c|^k, \qquad C_k^{\mathrm{act}} = \frac{3(8/3)^k}{1 - 2^{-k}}.$$

This includes an empty coarse group, $x = c$, the largest sampled level $j = J - 1$, and arbitrarily large $|x - c|$.

Proof. Fix c, x, put $R = |x - c|$, and define

$$\mathcal{A}_{\mathrm{c}}(R) := \left\{ j \in \mathcal{J}_{\mathrm{c}} : h_j < \frac{8R}{3} \right\}.$$

Multiplying the envelope in Lemma A.15 by $h_j^{k-1} > 0$ and summing gives

$$\sum_{j\in\mathcal{J}_c} h_j^{k-1}|\Delta_j^c(x)| \le 3 \sum_{j\in\mathcal{A}_c(R)} h_j^k. \tag{7.2}$$

If $R = 0$, the cutoff set is empty and the claim is $0 \le 0$. If $R > 0$, Lemma A.16, with $\alpha = k$ and $T = 8R/3$, gives

$$\sum_{j\in\mathcal{A}_c(R)} h_j^k < \frac{(8R/3)^k}{1-2^{-k}}.$$

Combining with (7.2),

$$\sum_{j\in\mathcal{J}_c} h_j^{k-1}|\Delta_j^c(x)| < \frac{3(8/3)^k}{1-2^{-k}} R^k = C_k^{\mathrm{act}} R^k,$$

which implies the stated non-strict bound. The denominator is positive because $k > 1$, so the constant is finite and depends only on k. Empty groups and cutoff sets are covered by the empty-sum convention.

No restriction $R \le H$ is used. An arbitrarily distant observation may make every sampled coarse scale eligible, including $j = J-1$, but those scales remain a finite subset below $8R/3$ and are charged once to R^k. At $R = 3h_j/8$, that scale is excluded by the strict cutoff and is inactive by Lemma A.14. □

Completion of the activation and support certificate. Lemma A.14 applies the scale-j and scale-$j+1$ cell margins to prove

$$|x-c| \le \frac{3h_j}{8} \quad\Longrightarrow\quad D_j^c(x) - D_j^c(c) = 0$$

for every sampled level, including equality. It applies the scale-J margin to the exact top residual to prove

$$|x-c| \le \frac{3H}{8} \quad\Longrightarrow\quad R_H^c(x) = 0.$$

The proof covers both displacement signs, phase endpoints, candidate-grid boundaries, $j = 0$, $j = J-1$, and the exact $x = c$ baseline supplied by Lemma A.13.

Lemma A.15 combines support with the strict digit range to prove

$$\left|D_j^c(x) - D_j^c(c)\right| < 3h_j, \qquad \left|D_j^c(x) - D_j^c(c)\right| \le 3h_j \mathbf{1}\left\{h_j < \frac{8|x-c|}{3}\right\}.$$

The strict cutoff retains inactivity at $|x-c| = 3h_j/8$. Lemma A.16 then controls accumulation without counting levels. Proposition A.8 gives

$$\sum_{j\in\mathcal{J}_f} \left|D_j^c(x) - D_j^c(c)\right| \le 16\min\{|x-c|, \sigma\},$$

and Proposition A.9 gives

$$\sum_{j\in\mathcal{J}_c} h_j^{k-1}\left|D_j^c(x) - D_j^c(c)\right| \le \frac{3(8/3)^k}{1-2^{-k}}|x-c|^k.$$

Empty fine or coarse groups contribute zero. The fine constant is independent of the number of fine levels, and the coarse estimate remains valid when a rare observation activates every available scale. These results jointly prove every stated conclusion and export exactly the activation/support certificate consumed by the fine-variance, coarse-variance, and top-bias steps. No moment expectation, variance conclusion, localization event, or future bias estimate is imported. □

A.8 Conditional Mean and Second Moment

Lemma A.17 (Exact level and offset law). *Under Assumption 3 and the setting-defined ranges $k > 1$, $0 < \epsilon < \sigma$, $\gamma_k \in (0, 1)$, and $b_k \geq 1$, the sampled-level set $\{0, \ldots, J - 1\}$ is nonempty. The fine and coarse sets partition it. For each nonempty group $G \in \{\mathcal{J}_{\mathrm{f}}, \mathcal{J}_{\mathrm{c}}\}$, $W_G \in (0, \infty)$, and every $j \in G$ satisfies*

$$p_j = \frac{1}{m}\frac{w_j}{W_G} > 0, \qquad \sum_{j \in G} p_j = \frac{1}{m}.$$

Consequently $\sum_{j=0}^{J-1} p_j = 1$. If one auxiliary group is empty, it contributes no level, has no normalizer used in any formula, and the other group has $m = 1$ and total mass one. Moreover, conditional on any fixed center t and any level j,

$$\Pr\{(A_i, B_i) = (a_j(t), a_{j+1}(t)) \mid L_i = j\} = \frac{1}{16} \quad \textit{for each fixed } t.$$

These conclusions include the smallest level $j = 0$, the top sampled level $j = J - 1$, and the case $J = 1$.

Proof. From the setting definitions,

$$\frac{H_*}{h_0} = \frac{b_k}{\gamma_k}\left(\frac{\sigma}{\epsilon}\right)^{k/(k-1)} > 1.$$

Thus $J = \lceil \log_2(H_*/h_0) \rceil \geq 1$, so the sampled-level set is finite and nonempty. Each level satisfies exactly one of $h_j \leq \sigma$ and $h_j > \sigma$, including equality in the fine group, so the two groups form a disjoint partition.

For every level, $h_j/\sigma > 0$. Hence both possible definitions of w_j are finite and strictly positive. A nonempty finite group therefore has $0 < W_G = \sum_{\ell \in G} w_\ell < \infty$, which proves $p_j > 0$ and

$$\sum_{j \in G} p_j = \frac{1}{m}\frac{\sum_{j \in G} w_j}{W_G} = \frac{1}{m}.$$

There are exactly $m \in \{1, 2\}$ nonempty groups, so summing the last identity over them gives total mass one. An empty group has no index in the level sum and the setting does not define or use its W_G, so no empty sum appears in a denominator.

Finally, A_i and B_i are independent and uniform on the four-element set $\mathcal{S}$, independently of L_i and the center. Once t and j are fixed, the selected values $a_j(t), a_{j+1}(t)$ are fixed members of $\mathcal{S}$. Their ordered pair therefore has probability $(1/4)^2 = 1/16$. At $j = J - 1$, the second selector is $a_J(t)$, which is setting-defined; when $J = 1$, this is the same sole sampled-level calculation with $j = 0 = J - 1$. □

Lemma A.18 (Refinement kernel and integrability). *Under Assumption 3, Proposition A.1 and Lemmas A.2 and A.3 provide the localization center. Lemma A.10 and Propositions A.5 and A.6 provide the strict digit range and precommitment conclusions. Together with Lemma A.17, let*

$$\mathscr{F}_{\mathrm{loc}} := \sigma(R_{\mathrm{loc}}, (X_r)_{r \in I_{\mathrm{loc}}}),$$

with the deterministic localization schedule understood as fixed. Then c is $\mathscr{F}_{\mathrm{loc}}$-measurable, and, for each $i \in I_{\mathrm{ref}}$,

$$\mathcal{L}\big(X_i, L_i, A_i, B_i, U_i \mid \mathscr{F}_{\mathrm{loc}}\big) = D \otimes (p_0, \dots, p_{J-1}) \otimes \mathrm{Unif}(\mathcal{S})^{\otimes 2} \otimes \mathrm{Unif}[-1, 2] \quad \textit{almost surely.}$$

For every deterministic $t \in \mathbb{R}$, $Z_i(t)$ is Borel measurable, integrable, and square-integrable. In fact,

$$|Z_i(t)| \le 48 \max_{0 \le j < J} \frac{h_j}{p_j} < \infty \quad \textit{pathwise.}$$

Also, for

$$\Delta_j^t(x) := D_j^t(x) - D_j^t(t),$$

one has $|\Delta_j^t(x)| < 3h_j$, so $T_t(X) = \sum_{j=0}^{J-1} \Delta_j^t(X)$ and every term in the proposed mean and raw-square formulas is integrable. The resulting right-hand sides are Borel functions of t.

Proof. The Proposition A.1, Lemma A.2, and Lemma A.3 midpoint result makes c a measurable function of the localization seed and localization transcript, and hence of $\mathscr{F}_{\mathrm{loc}}$. Assumption 3 makes every refinement sample and seed independent of all localization samples and seeds, and gives the displayed product law within the refinement tuple. Therefore conditioning on $\mathscr{F}_{\mathrm{loc}}$ changes none of the refinement marginals or their mutual independence. The response

$$Y_i = \mathbf{1}\{F_{L_i, A_i, B_i}(X_i)/h_{L_i} \ge U_i\}$$

is a Borel function of that tuple by Proposition A.5. It is not an independent seed, and the dither calculation below never conditions on Y_i.

The centered indicator difference in $Z_i(t)$ belongs to $\{-1, 0, 1\}$. Lemma A.17 gives $p_j > 0$ at each of finitely many levels. Hence

$$|Z_i(t)| \le \frac{16}{p_{L_i}} 3h_{L_i} \le 48 \max_{0 \le j < J} \frac{h_j}{p_j} < \infty.$$

This deterministic finite bound proves both integrability and square-integrability, conditionally and unconditionally, without a moment condition on D.

The digit range places each of $D_j^t(x)$ and $D_j^t(t)$ in $(-h_j, 2h_j)$. Their difference therefore lies in $(-3h_j, 3h_j)$, proving $|\Delta_j^t(x)| < 3h_j$. Since J is finite, $T_t(X)$, $\sum_j \mathbb{E}_D \Delta_j^t(X)$, and $\sum_j p_j^{-1} h_j \mathbb{E}_D |\Delta_j^t(X)|$ are all absolutely finite. Thus every finite sum and iterated expectation below is legal.

For measurability in t, the Borel selector and the finite offset set give the finite representation

$$D_j^t(x) = \sum_{a, b \in \mathcal{S}} \mathbf{1}\{(a_j(t), a_{j+1}(t)) = (a, b)\} F_{j,a,b}(x).$$

For each (j, a, b), the quantity $\mathbb{E}_D F_{j,a,b}(X)$ is constant as a function of t. The mean kernel is therefore a finite combination of Borel selector indicators, these constants, and the Borel value $D_j^t(t)$. For the absolute kernel, for fixed (j, a, b) define

$$\phi_{j,a,b}(z) := \mathbb{E}_D |F_{j,a,b}(X) - z|.$$

It is finite and satisfies $|\phi(z) - \phi(z')| \le |z - z'|$, so it is continuous. Hence the selected finite combination of $\phi_{j,a,b}(F_{j,a,b}(t))$ is Borel. This proves the last assertion without any interchange of an infinite sum or limit. $\square$

Proposition A.10 (Exact importance-weighted conditional mean)**.** *Under Assumption 3, Lemmas A.17 and A.18, and the exact dither first-moment identity from Lemma A.10, Proposition A.5, and Proposition A.6, define for every deterministic* $t \in \mathbb{R}$

$$\theta(t) := \sum_{j=0}^{J-1} \mathbb{E}_D\big[D_j^t(X) - D_j^t(t)\big] = \mathbb{E}_D T_t(X).$$

Then, for every $i \in I_{\mathrm{ref}}$*,*

$$\mathbb{E}[Z_i(c) \mid \mathscr{F}_{\mathrm{loc}}] = \theta(c) \quad \textit{almost surely,}$$

and consequently

$$\boxed{\mathbb{E}[Z_i(c) \mid c] = \theta(c) = \mathbb{E}_D T_c(X)} \quad \textit{almost surely.}$$

The equality is exact: the level probability p_j*, selected-pair probability* $1/16$*, and importance factor* $16/p_j$ *cancel without a remainder.*

Proof. First fix a deterministic center t and a level j. Use only in this proof

$$M_j^t := \mathbf{1}\{(A_i, B_i) = (a_j(t), a_{j+1}(t))\}.$$

On $\{L_i = j\}$, the definition of $Z_i(t)$ is

$$Z_i(t) = \frac{16}{p_j} M_j^t G_{j,A_i,B_i}^t(X_i, U_i),$$

where, for $a, b \in \mathcal{S}$,

$$G_{j,a,b}^t(x,u) := 3h_j \left[\mathbf{1}\left\{\frac{F_{j,a,b}(x)}{h_j} \ge u\right\} - \mathbf{1}\left\{\frac{F_{j,a,b}(t)}{h_j} \ge u\right\}\right].$$

Conditional on (X_i, L_i, A_i, B_i), Proposition A.6 applies because U_i remains uniform and independent. Only the single selected ordered pair survives multiplication by M_j^t. Since that pair has probability $1/16$, for every x,

$$\begin{aligned}
&\mathbb{E}_{A_i,B_i,U_i}\big[M_j^t G_{j,A_i,B_i}^t(x, U_i) \mid L_i = j\big] \\
&\qquad = \frac{1}{16}\big(D_j^t(x) - D_j^t(t)\big) = \frac{1}{16}\Delta_j^t(x).
\end{aligned}$$

Now average over $X_i \sim D$ and over the level law. Lemma A.18 justifies every iteration and finite sum:

$$\begin{aligned}
\mathbb{E} Z_i(t) &= \sum_{j=0}^{J-1} p_j \frac{16}{p_j}\frac{1}{16}\mathbb{E}_D \Delta_j^t(X) \\
&= \sum_{j=0}^{J-1} \mathbb{E}_D \Delta_j^t(X) = \mathbb{E}_D T_t(X) = \theta(t).
\end{aligned}$$

By Lemma A.18, the same product-law calculation is valid conditional on $\mathscr{F}_{\mathrm{loc}}$, with its realized center substituted for t. This gives $\mathbb{E}[Z_i(c) \mid \mathscr{F}_{\mathrm{loc}}] = \theta(c)$. Since $\sigma(c) \subseteq \mathscr{F}_{\mathrm{loc}}$ and $\theta(c)$ is $\sigma(c)$-measurable, the tower property gives

$$\mathbb{E}[Z_i(c) \mid c] = \mathbb{E}[\theta(c) \mid c] = \theta(c).$$

The proof does not condition on localization success; the identity holds on and off $\mathcal{E}_{\mathrm{loc}}$. ∎

Proposition A.11 (Exact raw second moment with retained square structure)**.** *Under Assumption 3, Lemmas A.17 and A.18, and the exact dither square from Lemma A.10, Proposition A.5, and Proposition A.6, for every $i \in I_{\mathrm{ref}}$,*

$$\boxed{\mathbb{E}[Z_i(c)^2 \mid c] = 48 \sum_{j=0}^{J-1} \frac{h_j}{p_j} \mathbb{E}_D \left| D_j^c(X) - D_j^c(c) \right|} \quad \textit{almost surely.}$$

The identical equality holds conditional on $\mathscr{F}_{\mathrm{loc}}$. More explicitly, for each nonempty group G, substitution of $p_j = m^{-1} w_j / W_G$ gives its exact contribution

$$48 m W_G \sum_{j \in G} \frac{h_j}{w_j} \mathbb{E}_D |D_j^c(X) - D_j^c(c)|.$$

Thus a nonempty fine group contributes exactly

$$48 m W_{\mathrm{f}} \sigma \sum_{j \in \mathcal{J}_{\mathrm{f}}} \mathbb{E}_D |D_j^c(X) - D_j^c(c)|,$$

and a nonempty coarse group contributes exactly

$$48 m W_{\mathrm{c}} \sigma^{2-k} \sum_{j \in \mathcal{J}_{\mathrm{c}}} h_j^{k-1} \mathbb{E}_D |D_j^c(X) - D_j^c(c)|.$$

An empty group contributes no term and invokes no W_G.

Proof. For fixed t, write $Z_i(t)$ as the exact disjoint-level sum

$$Z_i(t) = \sum_{j=0}^{J-1} \mathbf{1}\{L_i = j\} \frac{16}{p_j} M_j^t G_{j,A_i,B_i}^t (X_i, U_i).$$

Since $\mathbf{1}\{L_i = j\}\mathbf{1}\{L_i = \ell\} = 0$ whenever $j \neq \ell$, all cross-level products vanish pathwise, not by an expectation argument. Also $(M_j^t)^2 = M_j^t$. Therefore

$$Z_i(t)^2 = \sum_{j=0}^{J-1} \mathbf{1}\{L_i = j\} \frac{256}{p_j^2} M_j^t \big(G_{j,A_i,B_i}^t (X_i, U_i)\big)^2.$$

The internal centered-indicator cross term is not dropped. Indeed, with

$$I_x = \mathbf{1}\{F_{j,a,b}(x)/h_j \geq U_i\}, \qquad I_t = \mathbf{1}\{F_{j,a,b}(t)/h_j \geq U_i\},$$

one has exactly

$$(I_x - I_t)^2 = I_x + I_t - 2 I_x I_t.$$

The Lemma A.10, Proposition A.5, and Proposition A.6 calculation retains this cross term and identifies the square with the indicator of the half-open interval between the two thresholds. Consequently, for the selected pair,

$$\mathbb{E}_{U_i} \left[\big(G_{j,a_j(t),a_{j+1}(t)}^t (x, U_i)\big)^2 \right] = 3 h_j |\Delta_j^t(x)|.$$

This equality includes threshold ties, both endpoints of the dither support, reversed threshold order, and $x = t$. Averaging the match indicator over the independent uniform offsets contributes exactly $1/16$. Averaging next over X_i and the level law gives

$$
\begin{aligned}
\mathbb{E}Z_i(t)^2 &= \sum_{j=0}^{J-1} p_j \frac{256}{p_j^2} \frac{1}{16} 3h_j \mathbb{E}_D|\Delta_j^t(X)| \\
&= 48 \sum_{j=0}^{J-1} \frac{h_j}{p_j} \mathbb{E}_D|\Delta_j^t(X)|.
\end{aligned}
$$

This is a raw second moment: no subtraction of $\theta(t)^2$, domination, activity estimate, or variance upper bound has occurred. Lemma A.18 applies the deterministic-center calculation conditional on $\mathscr{F}_{\mathrm{loc}}$, and the tower property then yields the displayed conditioning on c.

Finally, if $j \in G$, then $p_j^{-1} = mW_G/w_j$. For fine levels, $h_j/w_j = \sigma$. For coarse levels,

$$
\frac{h_j}{w_j} = h_j \left(\frac{h_j}{\sigma}\right)^{k-2} = \sigma^{2-k} h_j^{k-1}.
$$

Substitution gives the two exact group expressions. No normalizer is evaluated for an empty group. □

Proposition A.12 (Exact telescope target, residuals, and zero baseline)**.** *Under Assumption 2, Lemma A.1, Proposition A.10, Proposition A.7, and Lemma A.13, for every deterministic $t \in \mathbb{R}$,*

$$
\theta(t) = \mathbb{E}_D T_t(X), \qquad (\mu - t) - \theta(t) = \mathbb{E}_D R_0^t(X) + \mathbb{E}_D R_H^t(X).
$$

Consequently, for the generated center,

$$
\boxed{(\mu - c) - \theta(c) = \mathbb{E}_D R_0^c(X) + \mathbb{E}_D R_H^c(X)} \quad \textit{almost surely.}
$$

Neither signed residual is bounded or discarded here. Moreover, for every realization of the seeds, $X_i = c$ implies

$$
Z_i(c) = 0, \qquad D_j^c(X_i) - D_j^c(c) = 0 \textit{ for all } j, \qquad T_c(c) = R_0^c(c) = R_H^c(c) = 0.
$$

Proof. Lemma A.1 and $|\mu| \le \lambda$ give

$$
\mathbb{E}_D|X| \le \mathbb{E}_D|X - \mu| + |\mu| \le \sigma + \lambda < \infty.
$$

Thus the integrability condition in Proposition A.7 is discharged. That proposition gives, for each deterministic t,

$$
(\mu - t) - \mathbb{E}_D T_t(X) = \mathbb{E}_D R_0^t(X) + \mathbb{E}_D R_H^t(X).
$$

Proposition A.10 identifies $\mathbb{E}_D T_t(X) = \theta(t)$, proving the two displayed residual formulas, first for every deterministic t and then at the measurable random value $t = c$. No interchange with the random center is required: the deterministic identity is simply evaluated at its realized value.

If $X_i = c$, then the response indicator and the decoder-centering indicator in the definition of $Z_i(c)$ have the same level, offsets, threshold, and dither, so their difference is zero for every U_i. Hence $Z_i(c) = 0$ pathwise, regardless of the level probability or whether the offset pair matches. Exact vanishing of every centered digit, the telescope, and both residuals is the zero-displacement conclusion from Lemma A.11, Lemma A.12, Proposition A.7, and Lemma A.13. □

Completion of the conditional-moment certificate. Lemma A.17 proves that all sampled-level probabilities are strictly positive, normalized exactly, and legal even when one auxiliary group is empty. It also supplies the exact ordered-pair probability $1/16$, including at $j = 0$, $j = J - 1$, and $J = 1$. Lemma A.18 then makes the center supplied by Lemma A.2 measurable with respect to $\mathscr{F}_{\mathrm{loc}}$, proves that the complete refinement tuple retains its product law after localization conditioning, and establishes conditional integrability and square-integrability before any expectation is interchanged.

Using that legal kernel and the exact dither first moment, Proposition A.10 tracks

$$p_j \times \frac{16}{p_j} \times \frac{1}{16} = 1$$

at every level and obtains the exact identity

$$\mathbb{E}[Z_i(c) \mid c] = \theta(c) = \sum_{j=0}^{J-1} \mathbb{E}_D[D_j^c(X) - D_j^c(c)] = \mathbb{E}_D T_c(X).$$

Using the exact dither square, Proposition A.11 tracks

$$p_j \times \frac{256}{p_j^2} \times \frac{1}{16} \times 3h_j = \frac{48h_j}{p_j}$$

and obtains

$$\mathbb{E}[Z_i(c)^2 \mid c] = 48 \sum_{j=0}^{J-1} \frac{h_j}{p_j} \mathbb{E}_D|D_j^c(X) - D_j^c(c)|.$$

Its derivation preserves the exact within-digit square/cross expression, and cross-level terms vanish only because one and only one level is sampled. The proposition also substitutes the exact fine and coarse laws without bounding a normalizer or importing a future activity argument.

Finally, Proposition A.12 uses two exact identities: the telescope in Lemma A.11 and the floor-remainder result in Lemma A.12. It combines them with Proposition A.7, Lemma A.13, and the exact mean to retain the complete same-target residual

$$(\mu - c) - \theta(c) = \mathbb{E}_D R_0^c(X) + \mathbb{E}_D R_H^c(X),$$

and proves the exact $X_i = c$ baseline. These results establish every stated conclusion without using a later activity, variance, or bias bound. □

A.9 Fine-Scale Conditional Variance

Lemma A.19 (Exact reduction of the fine conditional square)**.** *Under Assumptions 1 and 3, the exact group law in Proposition A.2, and the raw-square identity in Proposition A.11, define, for a deterministic center $t \in \mathbb{R}$, the appendix-local fine contribution*

$$\mathcal{V}_{\mathrm{f}}(t) := 48 \sum_{j \in \mathcal{J}_{\mathrm{f}}} \frac{h_j}{p_j} \mathbb{E}_D|D_j^t(X) - D_j^t(t)|,$$

where an empty fine sum is zero. If the fine group is nonempty, then

$$\boxed{\mathcal{V}_{\mathrm{f}}(t) = 48 m W_{\mathrm{f}} \sigma \, \mathbb{E}_D\left[\sum_{j \in \mathcal{J}_{\mathrm{f}}} |D_j^t(X) - D_j^t(t)|\right].} \tag{9.1}$$

For the generated center c, the conditional raw square splits almost surely as

$$\mathbb{E}[Z_i(c)^2 \mid c] = \mathcal{V}_{\mathrm{f}}(c) + 48 \sum_{j \in \mathcal{J}_{\mathrm{c}}} \frac{h_j}{p_j} \mathbb{E}_D |D_j^c(X) - D_j^c(c)|. \tag{9.2}$$

No normalizer is evaluated for an empty group.

Proof. If $\mathcal{J}_{\mathrm{f}} = \varnothing$, the definition of $\mathcal{V}_{\mathrm{f}}(t)$ is an empty sum and hence equals zero. There is then no fine index j, no fine probability formula to evaluate, and no division by $W_{\mathrm{f}} = 0$. This proves the empty-group clause.

Suppose $\mathcal{J}_{\mathrm{f}} \neq \varnothing$. The group law gives $m \in \{1, 2\}$, $W_{\mathrm{f}} > 0$, and, for every fine index,

$$w_j = \frac{h_j}{\sigma}, \qquad p_j = \frac{1}{m} \frac{w_j}{W_{\mathrm{f}}} = \frac{1}{m} \frac{h_j/\sigma}{W_{\mathrm{f}}} = \frac{h_j}{m\sigma W_{\mathrm{f}}}. \tag{9.3}$$

All divisions are legal because $h_j > 0$, $\sigma > 0$, $W_{\mathrm{f}} > 0$, and $m \geq 1$. Taking the reciprocal in (9.3) and multiplying by the same positive h_j gives the exact, level-independent identity

$$p_j^{-1} = \frac{m\sigma W_{\mathrm{f}}}{h_j}, \qquad \frac{h_j}{p_j} = mW_{\mathrm{f}}\sigma. \tag{9.4}$$

In particular, if $h_j = \sigma$, the setting assigns the level to the fine group, $w_j = 1$, and (9.3)–(9.4) remain unchanged.

Substitution of (9.4), with no inequality, into the exact factor-48 fine sum gives

$$\mathcal{V}_{\mathrm{f}}(t) = 48mW_{\mathrm{f}}\sigma \sum_{j \in \mathcal{J}_{\mathrm{f}}} \mathbb{E}_D |D_j^t(X) - D_j^t(t)|. \tag{9.5}$$

The level set is finite. Moreover, Lemma A.18 gives $|D_j^t(X) - D_j^t(t)| < 3h_j$ for every term. Finite linearity of expectation therefore yields the exact order exchange

$$\sum_{j \in \mathcal{J}_{\mathrm{f}}} \mathbb{E}_D |D_j^t(X) - D_j^t(t)| = \mathbb{E}_D \left[\sum_{j \in \mathcal{J}_{\mathrm{f}}} |D_j^t(X) - D_j^t(t)| \right], \tag{9.6}$$

with no infinite-series, limit, or conditional-expectation interchange. Combining (9.5) and (9.6) proves (9.1).

Finally, Proposition A.11 gives the exact full conditional identity for the generated center. Partitioning its finite level sum into the disjoint fine and coarse index sets yields (9.2). The first sub-sum is exactly $\mathcal{V}_{\mathrm{f}}(c)$ by definition; nothing is bounded, subtracted, or imported about the coarse sub-sum. □

Proposition A.13 (Count-free fine-square bound)**.** *Under Lemmas A.19 and A.6, and Proposition A.8, for every deterministic center $t \in \mathbb{R}$, every setting-admissible law D, and every finite setting scale family,*

$$\boxed{\mathcal{V}_{\mathrm{f}}(t) \leq 768mW_{\mathrm{f}}\sigma^2 \leq 3072\sigma^2.} \tag{9.7}$$

The same bound holds almost surely at the generated center $t = c$. The constant $3072 = 48{\cdot}16{\cdot}2{\cdot}2$ is universal: it has no dependence on $k, \lambda, \sigma, \epsilon, \delta, D, t, J, H$, or the number of fine levels. If the fine group is empty, or if all fine digits are inactive D-almost surely, then $\mathcal{V}_{\mathrm{f}}(t) = 0$.

Proof. The empty-fine case was proved in Lemma A.19, so suppose the fine group is nonempty. The pathwise ledger applies before any expectation:

$$\sum_{j\in\mathcal{J}_{\mathrm{f}}} |D_j^t(x) - D_j^t(t)| \le 16 \min\{|x-t|, \sigma\} \qquad \text{for every } x \in \mathbb{R}. \tag{9.8}$$

All quantities in (9.8) are nonnegative and integrable by the finite bounded digit interface. Taking $X \sim D$, then using only the pointwise scalar inequality $\min\{|X-t|, \sigma\} \le \sigma$, gives

$$\begin{aligned} \mathbb{E}_D\left[\sum_{j\in\mathcal{J}_{\mathrm{f}}} |D_j^t(X) - D_j^t(t)|\right] &\le 16\,\mathbb{E}_D \min\{|X-t|, \sigma\} \\ &\le 16\sigma. \end{aligned} \tag{9.9}$$

No moment of $X - t$ is used. In particular, neither localization nor the recentered k-moment is needed to pass from the first to the second line. Substituting (9.9) into the exact reduction (9.1) gives

$$\mathcal{V}_{\mathrm{f}}(t) \le 48 m W_{\mathrm{f}} \sigma(16\sigma) = 768 m W_{\mathrm{f}} \sigma^2. \tag{9.10}$$

The level law gives $m \le 2$, and the geometric normalizer calculation gives $W_{\mathrm{f}} \le 2$, independently of the cardinality of $\mathcal{J}_{\mathrm{f}}$. Therefore

$$768 m W_{\mathrm{f}} \sigma^2 \le 768 \cdot 2 \cdot 2\,\sigma^2 = 3072\sigma^2,$$

which proves (9.7). In the auxiliary fine-only configuration, $m = 1$ and the same argument gives the sharper bound $1536\sigma^2$. Under the theorem design, both groups occur, $m = 2$, and the universal displayed bound applies.

The entire calculation is valid for every deterministic t; the conditional kernel then permits evaluation at the generated c, without conditioning on localization success. If every fine digit difference is zero D-almost surely, the exact nonnegative sum defining $\mathcal{V}_{\mathrm{f}}(t)$ is zero. In particular, when D is concentrated at t, the exact zero-displacement identity makes every fine integrand zero. ◻

Proposition A.14 (Fine conditional-variance interface)**.** *Under Proposition A.13, the conditional kernel and raw square in Proposition A.11, and Assumption 3, define only for this proof*

$$Z_{i,\mathrm{f}}(t) := Z_i(t)\mathbf{1}\{L_i \in \mathcal{J}_{\mathrm{f}}\}.$$

Then, for the generated center c,

$$\mathbb{E}[Z_{i,\mathrm{f}}(c)^2 \mid c] = \mathcal{V}_{\mathrm{f}}(c), \qquad \boxed{\mathrm{Var}(Z_{i,\mathrm{f}}(c) \mid c) \le 3072\sigma^2} \quad \textit{almost surely.} \tag{9.11}$$

For the full pseudo-observation, the precise assembly-facing consequence is

$$\boxed{\mathrm{Var}(Z_i(c) \mid c) \le 3072\sigma^2 + 48 \sum_{j\in\mathcal{J}_{\mathrm{c}}} \frac{h_j}{p_j}\,\mathbb{E}_D|D_j^c(X) - D_j^c(c)|} \quad \textit{almost surely.} \tag{9.12}$$

The coarse term in (9.12) is the untouched exact raw-square sub-sum, not a coarse estimate.

Proof. The Lemma A.18, Proposition A.10, Proposition A.11, and Proposition A.12 disjoint-level square calculation remains valid after multiplication by $\mathbf{1}\{L_i \in \mathcal{J}_{\mathrm{f}}\}$: it simply retains the fine level indices. Hence

$$\mathbb{E}[Z_{i,\mathrm{f}}(c)^2 \mid c] = 48 \sum_{j\in\mathcal{J}_{\mathrm{f}}} \frac{h_j}{p_j}\,\mathbb{E}_D|D_j^c(X) - D_j^c(c)| = \mathcal{V}_{\mathrm{f}}(c).$$

The square-integrability of $Z_i(c)$ also gives conditional square-integrability of its fine restriction. Therefore, directly from the definition of conditional variance,

$$\mathrm{Var}(Z_{i,\mathrm{f}}(c) \mid c) = \mathbb{E}[Z_{i,\mathrm{f}}(c)^2 \mid c] - \big(\mathbb{E}[Z_{i,\mathrm{f}}(c) \mid c]\big)^2 \le \mathcal{V}_{\mathrm{f}}(c) \le 3072\sigma^2.$$

For the full variable, the same variance identity and nonnegativity of a conditional square give

$$\mathrm{Var}(Z_i(c) \mid c) \le \mathbb{E}[Z_i(c)^2 \mid c].$$

Insert the exact fine/coarse raw-square split (9.2) and then apply (9.7) only to the fine term. This proves (9.12). We do not assert that the full conditional variance equals the sum of the conditional variances of its fine- and coarse-level restrictions: although those level events are disjoint pathwise, their conditional means need not vanish, so such an additive variance identity is not available. The raw-second-moment upper bound is the valid interface.

If $X_i = c$, the exact baseline gives $Z_i(c) = 0$ pathwise for every level and seed, and hence also $Z_{i,\mathrm{f}}(c) = 0$. If the fine group is empty, $Z_{i,\mathrm{f}}(c) = 0$ identically and (9.11) reads $0 \le 0$. If the coarse group is empty, the last sum in (9.12) is absent, so the same $3072\sigma^2$ bound applies to the full conditional variance. □

Completion of the fine-variance certificate. Proposition A.11 supplies the exact factor-48 conditional raw square under the independent refinement kernel. Lemma A.19 combines it with the exact fine law from Proposition A.2 and explicitly checks

$$w_j = \frac{h_j}{\sigma}, \qquad p_j = \frac{h_j}{m\sigma W_{\mathrm{f}}}, \qquad \frac{h_j}{p_j} = mW_{\mathrm{f}}\sigma.$$

It then moves the finite level sum inside expectation exactly, before any inequality is applied. Proposition A.13 uses the pathwise result

$$\sum_{j\in\mathcal{J}_{\mathrm{f}}} |D_j^c(X) - D_j^c(c)| \le 16\min\{|X-c|, \sigma\} \le 16\sigma$$

and the bounds $m \le 2$, $W_{\mathrm{f}} \le 2$ to obtain

$$48mW_{\mathrm{f}}\sigma\,\mathbb{E}_D \sum_{j\in\mathcal{J}_{\mathrm{f}}} |D_j^c(X) - D_j^c(c)| \le 3072\sigma^2.$$

Thus arbitrarily many fine scales are summed pathwise and never counted. Proposition A.14 finally applies the conditional variance definition: the fine-restricted conditional variance is at most $3072\sigma^2$, while the same universal amount is the fine contribution to the valid full-variance raw-second-moment bound. The coarse sub-sum is retained exactly for its legal later producer; no coarse or future concentration conclusion is imported.

These named results prove the stated result with the explicit universal choice $C = 3072$. The statement is simultaneous over all deterministic centers and all setting-admissible laws, almost sure when evaluated at the generated center, uniform over every finite J, valid at $h_j = \sigma$, and empty-group safe. □

A.10 Coarse-Scale Conditional Variance

For a deterministic decoder value t, write

$$\Delta_j^t(x) := D_j^t(x) - D_j^t(t).$$

Let $\mathbb{E}_t$ and Var_t denote expectation and variance under the independent refinement product kernel with the decoder parameter fixed at t. Define

$$Z_{i,\mathrm{c}}(t) := Z_i(t)\mathbf{1}\{L_i \in \mathcal{J}_\mathrm{c}\}.$$

When $\mathcal{J}_\mathrm{c} \neq \varnothing$, set

$$S_\mathrm{c}(t) := 48 \sum_{j\in\mathcal{J}_\mathrm{c}} \frac{h_j}{p_j}\mathbb{E}_D|\Delta_j^t(X)|,$$

and set $S_\mathrm{c}(t) = 0$ when the coarse group is empty. Thus no inverse probability or normalizer is evaluated for an absent group.

Lemma A.20 (Coarse square and conditional variance)**.** *Under Proposition A.11 and Proposition A.2, for every deterministic center t, the independent-refinement conditional kernel satisfies*

$$\mathbb{E}_t[Z_{i,\mathrm{c}}(t)^2] = S_\mathrm{c}(t), \qquad \mathrm{Var}_t(Z_{i,\mathrm{c}}(t)) \le S_\mathrm{c}(t).$$

If $\mathcal{J}_\mathrm{c} \neq \varnothing$, then exactly

$$\boxed{S_\mathrm{c}(t) = 48mW_\mathrm{c}\sigma^{2-k}\,\mathbb{E}_D\left[\sum_{j\in\mathcal{J}_\mathrm{c}} h_j^{k-1}|\Delta_j^t(X)|\right].} \tag{10.1}$$

If $\mathcal{J}_\mathrm{c} = \varnothing$, then $Z_{i,\mathrm{c}}(t) = 0$ pathwise and $S_\mathrm{c}(t) = 0$, without evaluating W_c, w_j/W_c, or p_j^{-1} for an absent level.

Proof. By definition,

$$Z_{i,\mathrm{c}}(t)^2 = Z_i(t)^2\mathbf{1}\{L_i \in \mathcal{J}_\mathrm{c}\}.$$

The raw-square proof expands $Z_i(t)^2$ over the disjoint events $\{L_i = j\}$. Restricting that same exact expansion to $j \in \mathcal{J}_\mathrm{c}$ therefore gives

$$\mathbb{E}_t[Z_{i,\mathrm{c}}(t)^2] = 48\sum_{j\in\mathcal{J}_\mathrm{c}} \frac{h_j}{p_j}\mathbb{E}_D|\Delta_j^t(X)| = S_\mathrm{c}(t).$$

This is a raw conditional square. The direct conditional-variance identity gives

$$\mathrm{Var}_t(Z_{i,\mathrm{c}}(t)) = S_\mathrm{c}(t) - \mathbb{E}_t[Z_{i,\mathrm{c}}(t)]^2 \le S_\mathrm{c}(t).$$

Now suppose the coarse group is nonempty. For every $j \in \mathcal{J}_\mathrm{c}$, the definitions, in their exact order, are

$$w_j = \left(\frac{h_j}{\sigma}\right)^{2-k}, \qquad W_\mathrm{c} = \sum_{\ell\in\mathcal{J}_\mathrm{c}} w_\ell, \qquad p_j = \frac{1}{m}\frac{w_j}{W_\mathrm{c}}.$$

Thus

$$\frac{1}{p_j} = mW_{\mathrm{c}} \left(\frac{h_j}{\sigma}\right)^{k-2} = mW_{\mathrm{c}}\sigma^{2-k} h_j^{k-2},$$

and hence, without an inequality,

$$\begin{aligned} S_{\mathrm{c}}(t) &= 48 \sum_{j \in \mathcal{J}_{\mathrm{c}}} h_j \, mW_{\mathrm{c}}\sigma^{2-k} h_j^{k-2} \mathbb{E}_D |\Delta_j^t(X)| \\ &= 48 mW_{\mathrm{c}}\sigma^{2-k} \sum_{j \in \mathcal{J}_{\mathrm{c}}} h_j^{k-1} \mathbb{E}_D |\Delta_j^t(X)| \\ &= 48 mW_{\mathrm{c}}\sigma^{2-k} \, \mathbb{E}_D \left[\sum_{j \in \mathcal{J}_{\mathrm{c}}} h_j^{k-1} |\Delta_j^t(X)| \right]. \end{aligned}$$

The last equality moves expectation across a finite sum only. Lemma A.18, Proposition A.10, Proposition A.11, and Proposition A.12 already proves every summand integrable. At $k = 2$, this same calculation reads $w_j = 1$, $p_j = (mW_{\mathrm{c}})^{-1}$, and $h_j/p_j = mW_{\mathrm{c}} h_j$; no limiting interpretation is used.

If the coarse group is empty, the defining indicator of $Z_{i,\mathrm{c}}$ is zero and every coarse sum is empty. This proves the exact zero statements directly and completes the two cases. $\square$

Proposition A.15 (Coarse-square moment closure)**.** *Under Assumption 2, Lemma A.4, Proposition A.9, and Lemma A.20, fix a deterministic center t with $|t - \mu| \le 50\sigma$. Then*

$$S_{\mathrm{c}}(t) \le A_k W_{\mathrm{c}} \sigma^2, \qquad A_k := 96 C_k^{\mathrm{rec}} C_k^{\mathrm{act}}, \tag{10.2}$$

where the empty-coarse case means $S_{\mathrm{c}}(t) = W_{\mathrm{c}} = 0$ under the inequality-only empty-sum extension. The constant is explicitly

$$A_k = 96 \, [2^{k-1}(1 + 50^k)] \left[\frac{3(8/3)^k}{1 - 2^{-k}} \right],$$

which is finite and positive for every fixed $k > 1$.

Proof. If the coarse group is empty, the conclusion is the exact equality $0 = 0$, so suppose it is nonempty. The activity ledger is pointwise in x:

$$\sum_{j \in \mathcal{J}_{\mathrm{c}}} h_j^{k-1} |\Delta_j^t(x)| \le C_k^{\mathrm{act}} |x - t|^k \qquad \text{for every } x \in \mathbb{R}. \tag{10.3}$$

This is the required order of operations: all active scales are summed before any expectation is taken. Applying $\mathbb{E}_D$ to the nonnegative pointwise inequality gives

$$\mathbb{E}_D \left[\sum_{j \in \mathcal{J}_{\mathrm{c}}} h_j^{k-1} |\Delta_j^t(X)| \right] \le C_k^{\mathrm{act}} \mathbb{E}_D |X - t|^k. \tag{10.4}$$

Lemma A.4 concerns this same deterministic decoder center t, and the refinement kernel leaves the refinement draw with its population law D after localization conditioning. Therefore the localized-center condition and Assumption 2 give

$$\mathbb{E}_D |X - t|^k \le C_k^{\mathrm{rec}} \sigma^k. \tag{10.5}$$

Substituting (10.4)–(10.5) into the exact identity (10.1) yields

$$\begin{aligned} S_{\mathrm{c}}(t) &\le 48 m W_{\mathrm{c}} \sigma^{2-k} C_k^{\mathrm{act}} C_k^{\mathrm{rec}} \sigma^k \\ &= 48 m C_k^{\mathrm{act}} C_k^{\mathrm{rec}} W_{\mathrm{c}} \sigma^2 \\ &\le 96 C_k^{\mathrm{act}} C_k^{\mathrm{rec}} W_{\mathrm{c}} \sigma^2 = A_k W_{\mathrm{c}} \sigma^2. \end{aligned}$$

The final inequality is the displayed deterministic bound $m \le 2$; under the actual theorem design $m = 2$, so its coefficient is exactly 96. No level count, tail truncation, or term absorption occurs.

For rare $R = |X - t| \gg H$, every sampled coarse level may be active, but (10.3) remains valid because Lemma A.14, Lemma A.15, Proposition A.8, and Proposition A.9 imposed no condition $R \le H$. At $X = t$, its left and right sides are both zero. ☐

Lemma A.21 (Superquadratic coarse variance)**.** *Under Proposition A.15 and Proposition A.3, if $k > 2$ is fixed and $|t - \mu| \le 50\sigma$, then*

$$S_{\mathrm{c}}(t) \le C_{k,>}^{\mathrm{c}} \sigma^2, \qquad C_{k,>}^{\mathrm{c}} := \frac{A_k}{1 - 2^{2-k}} < \infty. \tag{10.6}$$

Consequently $\mathrm{Var}_t(Z_{i,\mathrm{c}}(t)) \le C_{k,>}^{\mathrm{c}} \sigma^2$*.*

Proof. In the fixed regime $k > 2$, the coarse normalizer bound is

$$W_{\mathrm{c}} \le \frac{1}{1 - 2^{2-k}}.$$

The denominator is positive because $2^{2-k} < 1$. Substitution into (10.2) gives (10.6) directly. The conditional-variance conclusion follows from Lemma A.20. The constant may grow as $k \downarrow 2$; no uniform limit in k is claimed or needed. ☐

Lemma A.22 (Critical coarse variance with exactly one logarithm)**.** *Under Proposition A.15 and Proposition A.3, set $k = 2$ exactly. For every $|t - \mu| \le 50\sigma$,*

$$S_{\mathrm{c}}(t) \le C_2^{\mathrm{c}} \sigma^2 \log \frac{\sigma}{\epsilon}, \tag{10.7}$$

where the numerical constant

$$C_2^{\mathrm{c}} := A_2 \left(\log_2(2b_2) + \frac{1}{\log 2} \right)$$

depends on no problem quantity. Consequently the same upper bound holds for $\mathrm{Var}_t(Z_{i,\mathrm{c}}(t))$*.*

Proof. At $k = 2$, the coarse weights are exactly $w_j = 1$, so

$$W_{\mathrm{c}} = \#\mathcal{J}_{\mathrm{c}}.$$

This is the sole coarse-level count. The normalizer proof, using the largest sampled scale $h_{J-1} = H/2$ and $H < 2b_2\sigma(\sigma/\epsilon)$, gives

$$W_{\mathrm{c}} < \log_2(2b_2) + \frac{\log(\sigma/\epsilon)}{\log 2}. \tag{10.8}$$

The design has $\sigma/\epsilon \ge e$, hence $\log(\sigma/\epsilon) \ge 1$. Therefore the additive constant in (10.8) is dominated by the target logarithm through the explicit inequality

$$\log_2(2b_2) + \frac{\log(\sigma/\epsilon)}{\log 2} \le \left(\log_2(2b_2) + \frac{1}{\log 2} \right) \log \frac{\sigma}{\epsilon}. \tag{10.9}$$

Combining (10.2) and (10.9) proves (10.7). There is no second count: before W_{c} is applied, Proposition A.9 at $k=2$ gives the pointwise geometric charge

$$\sum_{j\in\mathcal{J}_{\mathrm{c}}} h_j|\Delta_j^t(x)| \le C_2^{\mathrm{act}}|x-t|^2,$$

which contains no factor $\#\mathcal{J}_{\mathrm{c}}$. Thus (10.7) contains exactly the one logarithm carried by W_{c}. The proof sets $k=2$ from the start and does not take either one-sided limit. □

Lemma A.23 (Subquadratic coarse variance)**.** *Under Proposition A.15 and Proposition A.3, if fixed $1<k<2$ and $|t-\mu|\le 50\sigma$, then*

$$S_{\mathrm{c}}(t) \le C_{k,<}^{\mathrm{c}}\sigma^k H^{2-k}, \qquad C_{k,<}^{\mathrm{c}} := \frac{A_k}{2^{2-k}(1-2^{-(2-k)})} < \infty. \tag{10.10}$$

Consequently the same bound holds for $\mathrm{Var}_t(Z_{i,\mathrm{c}}(t))$*.*

Proof. In the fixed regime $1<k<2$, the last-term-dominated normalizer bound is

$$W_{\mathrm{c}} \le \frac{1}{1-2^{-(2-k)}}\left(\frac{H}{2\sigma}\right)^{2-k}.$$

Substituting this inequality into (10.2), with no change to H, gives

$$\begin{aligned} S_{\mathrm{c}}(t) &\le \frac{A_k}{1-2^{-(2-k)}}\sigma^2\left(\frac{H}{2\sigma}\right)^{2-k} \\ &= \frac{A_k}{2^{2-k}(1-2^{-(2-k)})}\sigma^{2-(2-k)}H^{2-k} \\ &= C_{k,<}^{\mathrm{c}}\sigma^k H^{2-k}. \end{aligned}$$

Both denominator factors are positive because $2-k\in(0,1)$. The constant is finite for each fixed k, although it may grow as $k\uparrow 2$. This is not replaced by a uniform-in-k limit or by the critical-regime formula. The activity ledger already includes $|X-t|\gg H$, so no tail beyond H is omitted. □

Proposition A.16 (Three-regime coarse conditional variance certificate)**.** *Under Assumption 2, Lemma A.20, Proposition A.15, and Lemmas A.21, A.22, and A.23, on the generated localization event*

$$\mathcal{E}_{\mathrm{loc}} = \{|c-\mu|\le 50\sigma\},$$

the actual decoder center satisfies, almost surely in its localization transcript,

$$\mathrm{Var}(Z_{i,\mathrm{c}}(c)\mid c) \le S_{\mathrm{c}}(c) \le \begin{cases} C_{k,>}^{\mathrm{c}}\sigma^2, & k>2, \\ C_2^{\mathrm{c}}\sigma^2\log(\sigma/\epsilon), & k=2, \\ C_{k,<}^{\mathrm{c}}\sigma^k H^{2-k}, & 1<k<2. \end{cases} \tag{10.11}$$

Moreover, $S_{\mathrm{c}}(c)$ is exactly the coarse nonnegative summand in $\mathbb{E}[Z_i(c)^2\mid c]$, not a replacement for the full raw square. If the coarse group is empty, or if every coarse digit has zero activity, then $S_{\mathrm{c}}(c)=0$ and $Z_{i,\mathrm{c}}(c)=0$ conditionally almost surely. In particular, at the exact baseline $X_i=c$, the coarse component is zero pathwise for every level, offset, and dither realization.

Proof. Fix any localization transcript in $\mathcal{E}_{\mathrm{loc}}$, and denote its realized center by $t = c$. Lemma A.18 says that conditional on this transcript the refinement kernel is still the original one, so Lemma A.20 applies to the actual conditional square. Lemma A.4 supplies the same-center moment needed by Proposition A.15. Exactly one of the three mutually exclusive cases holds. Lemma A.21, Lemma A.22, or Lemma A.23, respectively, gives (10.11).

For the exact raw-square interface, Proposition A.11 gives

$$\mathbb{E}[Z_i(c)^2 \mid c] = 48 \sum_{j\in\mathcal{J}_{\mathrm{f}}} \frac{h_j}{p_j}\mathbb{E}_D|D_j^c(X) - D_j^c(c)| + S_{\mathrm{c}}(c),$$

where an empty fine sum is zero. This line is only an exact decomposition; no fine bound is asserted or imported here. Thus the proposition precisely certifies the coarse contribution while distinguishing it from the variance and from the full raw square.

If the coarse group is empty, Lemma A.20 gives exact pathwise zero. If all coarse digit differences vanish D-almost surely, the exact nonnegative formula gives $S_{\mathrm{c}}(c) = 0$; the conditional square of $Z_{i,\mathrm{c}}(c)$ is then zero, so that component is conditionally almost surely zero. At $X_i = c$, the definition of the centered dither bracket gives the stronger pathwise identity before expectation. □

Completion of the coarse-variance certificate. Proposition A.11 supplies the exact factor-48 conditional raw square. Lemma A.20 restricts that identity to the coarse levels, distinguishes the genuine conditional variance from its raw-square upper bound, and performs the exact substitution

$$\frac{h_j}{p_j} = mW_{\mathrm{c}}\sigma^{2-k}h_j^{k-1} \qquad (j \in \mathcal{J}_{\mathrm{c}}).$$

Proposition A.9 then sums all coarse activity pointwise before expectation, including the rare-tail case $|X - c| \gg H$. Lemma A.4 integrates the resulting $|X - c|^k$ charge under exactly the generated localization interface. Proposition A.15 composes these inputs and proves the common bound

$$S_{\mathrm{c}}(c) \le A_k W_{\mathrm{c}}\sigma^2 \quad \text{on } \mathcal{E}_{\mathrm{loc}}.$$

The three normalizer formulas are then applied separately: Lemma A.21 gives the fixed-$k > 2$ $C_k\sigma^2$ bound; Lemma A.22 gives the exact $k = 2$ bound with precisely one $\log(\sigma/\epsilon)$; and Lemma A.23 gives the fixed-$1 < k < 2$ $C_k\sigma^k H^{2-k}$ bound. Proposition A.16 packages those mutually exclusive conclusions for the actual random center, with exact empty-group and zero-displacement reductions. This proves the stated result without importing a fine-variance estimate, concentration result, or bias statement. □

A.11 Residual Bias and Target Transfer

Lemma A.24 (Top residual pointwise tail envelope)**.** *Under Lemma A.12, Proposition A.7, and Lemma A.14, for every $c, x \in \mathbb{R}$,*

$$|R_H^c(x)| \le \frac{11}{3}|x - c|\mathbf{1}\left\{|x - c| > \frac{3H}{8}\right\}.$$

In particular, the right-hand side is zero at $x = c$ and at the support boundary $|x - c| = 3H/8$.

Proof. For this proof only, write

$$\rho_J^c(y) := y - Q_J^c(y).$$

Lemma A.12 gives $0 \le \rho_J^c(y) < H$ for every real y. Therefore

$$\begin{aligned} R_H^c(x) &= Q_J^c(x) - Q_J^c(c) \\ &= (x-c) - \big(\rho_J^c(x) - \rho_J^c(c)\big), \end{aligned}$$

and hence, with $R = |x-c|$,

$$|R_H^c(x)| \le R + |\rho_J^c(x) - \rho_J^c(c)| < R + H. \tag{11.1}$$

If $R \le 3H/8$, Lemma A.14 gives $R_H^c(x) = 0$, including equality and both displacement signs. If $R > 3H/8$, then $H < 8R/3$, so (11.1) yields

$$|R_H^c(x)| < R + \frac{8R}{3} = \frac{11R}{3}.$$

Combining the two cases and relaxing the strict active-tail inequality to a non-strict one proves the displayed supported envelope. At $R = 0$, this also agrees with the exact zero-displacement identity. □

Lemma A.25 (Localized top-residual bias)**.** *Under Assumptions 1 and 2, Lemmas A.4 and A.7, and Lemma A.24, every localization transcript in $\mathcal{E}_{\mathrm{loc}}$ satisfies*

$$|\mathbb{E}_D R_H^c(X)| \le \mathbb{E}_D |R_H^c(X)| \le \overline{C}_k^{\mathrm{tail}} \frac{\sigma^k}{H^{k-1}} \le \frac{\epsilon}{8},$$

where

$$\overline{C}_k^{\mathrm{tail}} = \frac{11}{3}\left(\frac{8}{3}\right)^{k-1} C_k^{\mathrm{rec}}.$$

Proof. Fix an arbitrary localization transcript in $\mathcal{E}_{\mathrm{loc}}$, so its actual decoder output c is fixed during $\mathbb{E}_D$. Put $R = |X-c|$. Since $k > 1$ and $H > 0$, on the event $R > 3H/8$,

$$R = \frac{R^k}{R^{k-1}} \le \frac{R^k}{(3H/8)^{k-1}} = \left(\frac{8}{3}\right)^{k-1} \frac{R^k}{H^{k-1}}.$$

Multiplying by the tail indicator and applying Lemma A.24 gives the pointwise estimate

$$|R_H^c(X)| \le \frac{11}{3}\left(\frac{8}{3}\right)^{k-1} \frac{|X-c|^k}{H^{k-1}}. \tag{11.2}$$

The right-hand side is integrable because Lemma A.4 gives, on the fixed generated event,

$$\mathbb{E}_D |X-c|^k \le C_k^{\mathrm{rec}} \sigma^k.$$

Taking expectations in (11.2) therefore yields

$$\begin{aligned} \mathbb{E}_D |R_H^c(X)| &\le \frac{11}{3}\left(\frac{8}{3}\right)^{k-1} \frac{\mathbb{E}_D |X-c|^k}{H^{k-1}} \\ &\le \frac{11}{3}\left(\frac{8}{3}\right)^{k-1} C_k^{\mathrm{rec}} \frac{\sigma^k}{H^{k-1}} \\ &= \overline{C}_k^{\mathrm{tail}} \frac{\sigma^k}{H^{k-1}} \\ &\le \frac{\epsilon}{8}. \end{aligned}$$

The last line is exactly the endpoint calibration. Absolute integrability also gives

$$|\mathbb{E}_D R_H^c(X)| \le \mathbb{E}_D |R_H^c(X)|.$$

The calculation is uniform over every transcript in $\mathcal{E}_{\mathrm{loc}}$ and uses the moment about its actual center. $\square$

Lemma A.26 (Bottom quantization residual bias)**.** *Under Lemma A.12 and Lemma A.7, for every real center c and every law D,*

$$|\mathbb{E}_D R_0^c(X)| \le \mathbb{E}_D |R_0^c(X)| < h_0 = \frac{\epsilon}{8},$$

and consequently $|\mathbb{E}_D R_0^c(X)| \le \epsilon/8$.

Proof. Lemma A.12 gives the pointwise strict bound

$$|R_0^c(x)| < h_0 \qquad \text{for every } x \in \mathbb{R}.$$

Thus $R_0^c(X)$ is integrable for every law D, and monotonicity of expectation gives

$$|\mathbb{E}_D R_0^c(X)| \le \mathbb{E}_D |R_0^c(X)| < h_0.$$

Lemma A.7 identifies the same setting scale as $h_0 = \epsilon/8$. This proof does not require localization, a support restriction, or a moment bound. At $X = c$, the exact zero-displacement identity makes the residual zero rather than merely smaller than h_0. $\square$

Proposition A.17 (Localized telescope-to-mean bias certificate)**.** *Under Assumptions 1 and 2, Proposition A.12, and Lemmas A.25 and A.26, every localization transcript in $\mathcal{E}_{\mathrm{loc}}$ satisfies*

$$|\mathbb{E}_D R_0^c(X)| \le \frac{\epsilon}{8}, \qquad |\mathbb{E}_D R_H^c(X)| \le \frac{\epsilon}{8},$$

and

$$\boxed{|(\mu - c) - \theta(c)| \le \frac{\epsilon}{4}}.$$

Proof. Fix any localization transcript in $\mathcal{E}_{\mathrm{loc}}$. Proposition A.12 uses the same actual center throughout and gives the exact signed identity

$$(\mu - c) - \theta(c) = \mathbb{E}_D R_0^c(X) + \mathbb{E}_D R_H^c(X). \tag{11.3}$$

Lemma A.26 supplies the first component bound, and Lemma A.25 supplies the second. Applying the scalar triangle inequality to (11.3) yields

$$\begin{aligned} |(\mu - c) - \theta(c)| &\le |\mathbb{E}_D R_0^c(X)| + |\mathbb{E}_D R_H^c(X)| \\ &\le \frac{\epsilon}{8} + \frac{\epsilon}{8} = \frac{\epsilon}{4}. \end{aligned}$$

Both residuals remain separately visible, so cancellation is not needed for the bound. This proves the claimed estimate on the generated localization event. $\square$

Completion of the bias certificate. Lemma A.12 and Lemma A.14 concern the exact selected top quantizer at the actual decoder center. Lemma A.24 combines them to prove

$$|R_H^c(x)| \le \frac{11}{3}|x - c| \mathbf{1}\left\{|x - c| > \frac{3H}{8}\right\}.$$

Lemma A.4 then controls the k-moment about that same actual center on $\mathcal{E}_{\mathrm{loc}}$. Lemma A.25 composes the supported envelope and the elementary tail-power inequality to obtain

$$|\mathbb{E}_D R_H^c(X)| \le \overline{C}_k^{\mathrm{tail}} \frac{\sigma^k}{H^{k-1}} \le \frac{\epsilon}{8},$$

where the last inequality is precisely Lemma A.7.

Independently, Lemma A.12 gives $|R_0^c(x)| < h_0$, and Lemma A.26 combines this with the identity $h_0 = \epsilon/8$ to prove

$$|\mathbb{E}_D R_0^c(X)| \le \frac{\epsilon}{8}.$$

Finally, Proposition A.12 identifies $\theta(c)$ as the exact conditional pseudo-observation mean and supplies

$$(\mu - c) - \theta(c) = \mathbb{E}_D R_0^c(X) + \mathbb{E}_D R_H^c(X).$$

Proposition A.17 applies the triangle inequality with the two separately proved $\epsilon/8$ bounds and concludes

$$|(\mu - c) - \theta(c)| \le \frac{\epsilon}{4}.$$

This is exactly the claim of Proposition A.17, in absolute value, uniformly for every localized center and with no unaccounted residual, surrogate center, support assumption, confidence conversion, or changed rate interface. □

A.12 Conditional Median-of-Means

The constants entering the full conditional variance certificate are

$$C_k^{\mathrm{rec}} = 2^{k-1}(1 + 50^k), \qquad C_k^{\mathrm{act}} = \frac{3(8/3)^k}{1 - 2^{-k}}, \qquad A_k = 96 C_k^{\mathrm{rec}} C_k^{\mathrm{act}},$$

$$C_{k,>}^{\mathrm{c}} = \frac{A_k}{1 - 2^{2-k}} \quad (k > 2),$$

$$C_2^{\mathrm{c}} = A_2 \left(\log_2(2b_2) + \frac{1}{\log 2} \right) \quad (k = 2),$$

and

$$C_{k,<}^{\mathrm{c}} = \frac{A_k}{2^{2-k}(1 - 2^{-(2-k)})} \quad (1 < k < 2).$$

Define

$$\boxed{V_k := 3072\sigma^2 \mathbf{1}\{\mathcal{J}_{\mathrm{f}} \ne \varnothing\} + \mathbf{1}\{\mathcal{J}_{\mathrm{c}} \ne \varnothing\} \begin{cases} C_{k,>}^{\mathrm{c}} \sigma^2, & k > 2, \\ C_2^{\mathrm{c}} \sigma^2 \log(\sigma/\epsilon), & k = 2, \\ C_{k,<}^{\mathrm{c}} \sigma^k H^{2-k}, & 1 < k < 2. \end{cases}} \tag{12.1}$$

At least one group is nonempty because $J \ge 1$, and every applicable coefficient is positive and finite. Hence $0 < V_k < \infty$; an absent group contributes exactly zero.

For later use, the Bernoulli form of Hoeffding's inequality proved in Proposition A.19 is

$$\Pr \left\{ \frac{1}{q} \sum_{g=1}^{q} B_g - p \ge a \right\} \le e^{-2qa^2} \tag{12.2}$$

for iid Bernoulli variables of mean p and every $a > 0$.

Lemma A.27 (Transcript-conditional iid full variance certificate)**.** *Under Assumptions 1 and 3, Lemma A.18, Propositions A.10 and A.11, Proposition A.14, and Proposition A.16, let*

$$\mathscr{T}_{\mathrm{loc}} := \sigma(R_{\mathrm{loc}}, (Y_r)_{r\in I_{\mathrm{loc}}})$$

be the complete observable localization transcript. Then c and $\mathcal{E}_{\mathrm{loc}} = \{|c-\mu| \le 50\sigma\}$ are $\mathscr{T}_{\mathrm{loc}}$-measurable. Conditional on $\mathscr{T}_{\mathrm{loc}}$, the family $(Z_i(c))_{i\in I_{\mathrm{ref}}}$ is iid over all refinement samples, levels, offsets, and dithers, and

$$\mathbb{E}[Z_i(c) \mid \mathscr{T}_{\mathrm{loc}}] = \theta(c). \tag{12.3}$$

Uniformly for every transcript in $\mathcal{E}_{\mathrm{loc}}$,

$$\boxed{\mathrm{Var}(Z_i(c) \mid \mathscr{T}_{\mathrm{loc}}) \le V_k} \tag{12.4}$$

with V_k exactly as in (12.1). If a fine or coarse group is empty, its contribution in (12.1) and in the underlying exact raw square is zero. If the actual conditional variance is zero, then $Z_i(c) = \theta(c)$ conditionally almost surely.

Proof. By precommitment, each localization response Y_r is a measurable function of R_{loc} and X_r. Hence

$$\mathscr{T}_{\mathrm{loc}} \subseteq \mathscr{F}_{\mathrm{loc}},$$

where $\mathscr{F}_{\mathrm{loc}}$ is the localization sigma-field defined in Lemma A.18. The decoder definition makes c measurable with respect to $\mathscr{T}_{\mathrm{loc}}$, and therefore the event $\{|c-\mu| \le 50\sigma\}$ is also $\mathscr{T}_{\mathrm{loc}}$-measurable.

For $i \in I_{\mathrm{ref}}$, put

$$\Xi_i = (X_i, L_i, A_i, B_i, U_i).$$

Lemma A.18, together with the primitive mutual independence across refinement indices, gives for every finite collection of bounded Borel functions f_i,

$$\mathbb{E}\left[\prod_i f_i(\Xi_i) \,\middle|\, \mathscr{F}_{\mathrm{loc}}\right] = \prod_i \mathbb{E}[f_i(\Xi_i)] \quad \text{almost surely.}$$

Taking conditional expectation onto $\mathscr{T}_{\mathrm{loc}} \subseteq \mathscr{F}_{\mathrm{loc}}$ preserves this identity. Thus the complete refinement tuples are iid conditional on the observable transcript. Once the transcript is fixed, c is fixed, and $Z_i(c)$ is the same Borel function of Ξ_i for each index. Therefore the $Z_i(c)$ are conditionally iid. The exact pointwise mean calculation from Proposition A.10 then gives (12.3).

For a fixed transcript in $\mathcal{E}_{\mathrm{loc}}$, the raw-square decomposition gives

$$\begin{aligned} \mathrm{Var}(Z_i(c) \mid \mathscr{T}_{\mathrm{loc}}) &\le \mathbb{E}[Z_i(c)^2 \mid \mathscr{T}_{\mathrm{loc}}] \\ &= 48 \sum_{j\in\mathcal{J}_{\mathrm{f}}} \frac{h_j}{p_j} \mathbb{E}_D |D_j^c(X) - D_j^c(c)| \\ &\quad + 48 \sum_{j\in\mathcal{J}_{\mathrm{c}}} \frac{h_j}{p_j} \mathbb{E}_D |D_j^c(X) - D_j^c(c)|. \end{aligned} \tag{12.5}$$

This is an exact split of one nonnegative raw second moment, not a claim that the two component variances add. Proposition A.14 bounds the first sum by $3072\sigma^2$ when the fine group is nonempty and identifies it as zero when that group is empty. Proposition A.16 bounds the second sum by the applicable coarse term in (12.1) on the localized transcript and identifies it as zero when the coarse group is empty. Substitution into (12.5) proves (12.4) in all three mutually exclusive k-regimes, with exactly the group indicators in (12.1).

If the actual conditional variance is zero, then its definition together with (12.3) gives

$$\mathbb{E}[(Z_i(c) - \theta(c))^2 \mid \mathscr{T}_{\mathrm{loc}}] = 0.$$

A nonnegative random variable with zero conditional expectation is zero conditionally almost surely. This establishes the exact zero-variance conclusion without dividing by the variance. $\square$

Proposition A.18 (Exact block size and one-block success). *Under Lemma A.27, choose exactly*

$$s = \left\lceil \frac{32V_k}{\epsilon^2} \right\rceil$$

and use the fixed disjoint blocks $G_1, \ldots, G_q$ of size s from the setting. Then, for every g, uniformly on $\mathcal{E}_{\mathrm{loc}}$,

$$\boxed{\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}} \Pr\left\{ \left|\overline{Z}_g(c) - \theta(c)\right| > \frac{\epsilon}{2} \;\middle|\; \mathscr{T}_{\mathrm{loc}} \right\} \le \frac{1}{8} \mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}} \quad \textit{almost surely.} \tag{12.6}$$

If $\mathrm{Var}(Z_i(c) \mid \mathscr{T}_{\mathrm{loc}}) = 0$*, then the conditional probability in (12.6) is exactly zero.*

Proof. Fix a localization transcript in $\mathcal{E}_{\mathrm{loc}}$. By Lemma A.27, the s variables in G_g are conditionally iid with mean $\theta(c)$ and conditional variance at most V_k. Conditional independence gives

$$\mathbb{E}[\overline{Z}_g(c) \mid \mathscr{T}_{\mathrm{loc}}] = \theta(c)$$

and

$$\mathrm{Var}(\overline{Z}_g(c) \mid \mathscr{T}_{\mathrm{loc}}) = \frac{1}{s^2} \sum_{i \in G_g} \mathrm{Var}(Z_i(c) \mid \mathscr{T}_{\mathrm{loc}}) \le \frac{V_k}{s}. \tag{12.7}$$

If the variance in (12.7) is zero, the block mean equals $\theta(c)$ conditionally almost surely, so its bad probability is zero. Otherwise, conditional Chebyshev's inequality and the exact ceiling give

$$\begin{aligned}
\Pr\left\{ \left|\overline{Z}_g(c) - \theta(c)\right| > \frac{\epsilon}{2} \;\middle|\; \mathscr{T}_{\mathrm{loc}} \right\} &\le \frac{4\,\mathrm{Var}(\overline{Z}_g(c) \mid \mathscr{T}_{\mathrm{loc}})}{\epsilon^2} \\
&\le \frac{4V_k}{s\epsilon^2} \\
&\le \frac{4V_k}{(32V_k/\epsilon^2)\epsilon^2} \\
&= \frac{1}{8}.
\end{aligned} \tag{12.8}$$

Here $V_k > 0$ was proved after (12.1), so the last ratio is legal. Multiplying the pointwise transcript inequality by the $\mathscr{T}_{\mathrm{loc}}$-measurable indicator of $\mathcal{E}_{\mathrm{loc}}$ gives (12.6). No probability of the localization event has been used. $\square$

Proposition A.19 (Fixed odd-median amplification). *Under Lemma A.27 and Proposition A.18, choose exactly*

$$q = 2\left\lceil 8 \log \frac{4}{\delta} \right\rceil + 1.$$

Let $M(c)$ be the $(q+1)/2$-th order statistic of the fixed block means $\overline{Z}_1(c), \ldots, \overline{Z}_q(c)$; this is the median in the setting-defined decoder. Then, uniformly on $\mathcal{E}_{\mathrm{loc}}$,

$$\boxed{\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}} \Pr\left\{|M(c)-\theta(c)| > \frac{\epsilon}{2} \;\middle|\; \mathscr{T}_{\mathrm{loc}}\right\} \le \frac{\delta}{4}\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}} \quad \textit{almost surely.} \tag{12.9}$$

The order-statistic definition is fixed even when block values tie. Because q is odd, no choice between two central order statistics occurs.

Proof. Fix a localization transcript in $\mathcal{E}_{\mathrm{loc}}$, and define the proof-local indicators

$$B_g := \mathbf{1}\left\{|\overline{Z}_g(c) - \theta(c)| > \frac{\epsilon}{2}\right\}.$$

The blocks are fixed, disjoint, and have equal size. By Lemma A.27, they are conditionally iid; hence the B_g are conditionally iid Bernoulli with a common parameter $p \le 1/8$ by Proposition A.18.

Write $q = 2r+1$. If at most r block means are bad, at least $r+1$ belong to the closed interval

$$\left[\theta(c) - \frac{\epsilon}{2}, \theta(c) + \frac{\epsilon}{2}\right].$$

The $(r+1)$-th order statistic must then belong to the same interval. Equivalently,

$$\left\{|M(c)-\theta(c)| > \frac{\epsilon}{2}\right\} \subseteq \left\{\sum_{g=1}^{q} B_g \ge r+1\right\} \subseteq \left\{\frac{1}{q}\sum_{g=1}^{q} B_g \ge \frac{1}{2}\right\}. \tag{12.10}$$

This argument includes repeated block values and equality at either endpoint: endpoint equality is good because the bad event is strict.

For completeness, fix the conditional Bernoulli parameter p. For $t \ge 0$, let

$$\psi_p(t) = \log(1 - p + pe^t) - pt.$$

Then $\psi_p(0) = \psi_p'(0) = 0$, while

$$\psi_p''(t) = u_t(1-u_t) \le \frac{1}{4}, \qquad u_t = \frac{pe^t}{1-p+pe^t}.$$

Twice integrating the second-derivative bound gives $\psi_p(t) \le t^2/8$. Thus, for every $a > 0$, exponential Markov and conditional independence yield

$$\begin{aligned}\Pr\left\{\sum_{g=1}^{q}(B_g - p) \ge qa \;\middle|\; \mathscr{T}_{\mathrm{loc}}\right\} &\le \inf_{t>0} \exp\left(-tqa + \frac{qt^2}{8}\right) \\ &= \exp(-2qa^2),\end{aligned} \tag{12.11}$$

where the infimum is attained at $t = 4a$. Apply (12.11) with $a = 1/2 - p \ge 3/8$. From (12.10),

$$\Pr\left\{|M(c)-\theta(c)| > \frac{\epsilon}{2} \;\middle|\; \mathscr{T}_{\mathrm{loc}}\right\} \le \exp\left(-2q(1/2-p)^2\right) \le e^{-9q/32} \le e^{-q/4}. \tag{12.12}$$

The exact ceiling in q gives

$$q \ge 16 \log \frac{4}{\delta} + 1,$$

and hence

$$e^{-q/4} \leq \exp\left(-4\log\frac{4}{\delta}\right) = \left(\frac{\delta}{4}\right)^4 \leq \frac{\delta}{4}, \tag{12.13}$$

because $0 < \delta/4 < 1$. Combining (12.12)–(12.13) and multiplying by $\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}$ proves (12.9). This is a fixed finite majority calculation; no horizon or probability-mode upgrade is made. □

Proposition A.20 (Conditional exact-target refinement certificate)**.** *Under Assumptions 1 and 3, Proposition A.19, and the bias certificate in Proposition A.17, define the final setting estimator*

$$\widehat{\mu} = c + M(c).$$

Then, uniformly on every localization transcript in $\mathcal{E}_{\mathrm{loc}}$,

$$\Pr\left\{|\widehat{\mu} - \mu| > \frac{3\epsilon}{4} \;\middle|\; \mathscr{T}_{\mathrm{loc}}\right\} \leq \frac{\delta}{4}. \tag{12.14}$$

Equivalently, the indicator-valued conditional interface is

$$\boxed{\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}} \Pr\{|\widehat{\mu} - \mu| > \epsilon \mid \mathscr{T}_{\mathrm{loc}}\} \leq \frac{\delta}{4}\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}} \quad \textit{almost surely.} \tag{12.15}$$

This is the conditional interface used in the unconditional confidence argument; no expectation of (12.15) and no control of $\mathcal{E}_{\mathrm{loc}}^c$ is taken here.

Proof. Fix a localization transcript in $\mathcal{E}_{\mathrm{loc}}$. On the median-good event, Proposition A.17 is used once, in the single triangle inequality

$$\begin{aligned} |\widehat{\mu} - \mu| &= |M(c) - (\mu - c)| \\ &\leq |M(c) - \theta(c)| + |\theta(c) - (\mu - c)| \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{4} = \frac{3\epsilon}{4} < \epsilon. \end{aligned} \tag{12.16}$$

Therefore

$$\left\{|\widehat{\mu} - \mu| > \frac{3\epsilon}{4}\right\} \subseteq \left\{|M(c) - \theta(c)| > \frac{\epsilon}{2}\right\}$$

on that transcript. Proposition A.19 gives (12.14). Since $3\epsilon/4 < \epsilon$,

$$\{|\widehat{\mu} - \mu| > \epsilon\} \subseteq \{|\widehat{\mu} - \mu| > 3\epsilon/4\},$$

so multiplication by the transcript-measurable localization indicator gives (12.15). No localization failure probability is inserted or union-bounded. □

Completion of the conditional refinement certificate. Lemma A.27 first transfers the independent refinement kernel of Lemma A.18 from the larger localization sigma-field to the complete observable localization transcript. It proves conditional iid jointly over every refinement sample, level, offset, and dither, and retains the exact mean $\theta(c) = \mathbb{E}_D T_c(X)$.

The same lemma composes the interfaces at their legal raw-second-moment boundary. Proposition A.14 contributes $3072\sigma^2$ exactly when the fine group is present, while Proposition A.16 contributes exactly the applicable coarse certificate when the coarse group is present. This yields the full additive certificate (12.1):

$$\mathrm{Var}(Z_i(c) \mid \mathscr{T}_{\mathrm{loc}}) \leq V_k \quad \text{uniformly on } \mathcal{E}_{\mathrm{loc}}.$$

No additive decomposition of component variances is asserted. Empty groups are removed by explicit indicators, and zero actual variance gives exact conditional equality $Z_i(c) = \theta(c)$.

Proposition A.18 uses the specified choice

$$s = \lceil 32V_k/\epsilon^2 \rceil$$

and conditional block independence to obtain

$$\Pr\{|\overline{Z}_g(c) - \theta(c)| > \epsilon/2 \mid \mathscr{T}_{\mathrm{loc}}\} \le 1/8$$

for every fixed block on every localized transcript. Proposition A.19 uses the specified odd choice

$$q = 2\lceil 8\log(4/\delta)\rceil + 1$$

and the directly proved Hoeffding bound to show that a bad majority, and hence a median miss by more than $\epsilon/2$, has conditional probability at most $\delta/4$. The $(q+1)/2$-th order-statistic convention handles all repeated-value ties.

Finally, Proposition A.20 invokes the certificate in Proposition A.17, $|(\mu - c) - \theta(c)| \le \epsilon/4$ exactly once. It obtains total error $\epsilon/2 + \epsilon/4 = 3\epsilon/4 < \epsilon$ and exports the indicator-valued conditional inequality (12.15).

The allocation produced by this step is exactly

$$\boxed{N_{\mathrm{ref}} = qs = \left(2\left\lceil 8\log\frac{4}{\delta}\right\rceil + 1\right)\left\lceil\frac{32V_k}{\epsilon^2}\right\rceil.} \tag{12.17}$$

It is determined by known parameters and fixed design constants before any response is observed. Equations (12.1), (12.6), (12.9), (12.15), and (12.17) prove the stated result. The next two subsections establish unconditional confidence and the public sample bound. □

A.13 Unconditional Confidence

For a fixed $D \in \mathcal{D}(k, \lambda, \sigma)$, put

$$\mathcal{F}_D := \{|\widehat{\mu} - \mu(D)| > \epsilon\}. \tag{13.5}$$

The conditional refinement certificate and localization guarantee available at this point are, respectively,

$$\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}\mathbb{P}_D(\mathcal{F}_D \mid \mathscr{T}_{\mathrm{loc}}) \le \frac{\delta}{4}\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}} \quad \text{almost surely,} \tag{13.1}$$

and

$$\mathbb{P}_D(\mathcal{E}_{\mathrm{loc}}) \ge 1 - \frac{\delta}{4}, \qquad \mathcal{E}_{\mathrm{loc}} = \{|c - \mu(D)| \le 50\sigma\}. \tag{13.2}$$

We will use the defining conditional-expectation identity

$$\mathbb{E}[\mathbf{1}_A G] = \mathbb{E}[\mathbf{1}_A \mathbb{E}(G \mid \mathscr{T})] \tag{13.3}$$

for integrable G, $A \in \mathscr{T}$, and the elementary decomposition

$$\mathbb{P}(F) = \mathbb{P}(F \cap E) + \mathbb{P}(F \cap E^c) \le \mathbb{P}(F \cap E) + \mathbb{P}(E^c). \tag{13.4}$$

Lemma A.28 (Transcript tower identity)**.** *Under Assumptions 1, 2, and 3, Lemma A.2, and Proposition A.20,* $\mathcal{E}_{\mathrm{loc}} \in \mathscr{T}_{\mathrm{loc}}$*,* $\mathcal{F}_D$ *is measurable under the joint law* $\mathbb{P}_D$*, and*

$$\boxed{\mathbb{P}_D(\mathcal{F}_D \cap \mathcal{E}_{\mathrm{loc}}) = \mathbb{E}_D^{\mathrm{joint}}[\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}\mathbb{P}_D(\mathcal{F}_D \mid \mathscr{T}_{\mathrm{loc}})] .} \tag{13.6}$$

Here the conditional probability is any version of $\mathbb{E}_D^{\mathrm{joint}}[\mathbf{1}_{\mathcal{F}_D} \mid \mathscr{T}_{\mathrm{loc}}]$.

Proof. Lemma A.2 makes c a measurable function of the complete observable transcript. Since $\mu(D)$ and 50σ are fixed scalars under $\mathbb{P}_D$, $\mathcal{E}_{\mathrm{loc}} = \{|c-\mu(D)| \le 50\sigma\}$ belongs to $\mathscr{T}_{\mathrm{loc}}$.

Proposition A.20 uses the setting-defined variables $Z_i(c)$, finite block averages, and the fixed odd order-statistic median to define exactly

$$\widehat{\mu} = c + \operatorname{median}\big(\overline{Z}_1(c), \ldots, \overline{Z}_q(c)\big).$$

Those are finite measurable operations on the localization transcript, the refinement samples, and the refinement seeds. Hence $\widehat{\mu}$, and therefore $\mathcal{F}_D$, is measurable under the full joint law. In particular $\mathbf{1}_{\mathcal{F}_D}$ is integrable.

Because $\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}$ is $\mathscr{T}_{\mathrm{loc}}$-measurable, the defining property of conditional expectation gives

$$\begin{aligned} \mathbb{P}_D(\mathcal{F}_D \cap \mathcal{E}_{\mathrm{loc}}) &= \mathbb{E}_D^{\mathrm{joint}}[\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}\mathbf{1}_{\mathcal{F}_D}] \\ &= \mathbb{E}_D^{\mathrm{joint}}\Big[\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}\mathbb{E}_D^{\mathrm{joint}}[\mathbf{1}_{\mathcal{F}_D} \mid \mathscr{T}_{\mathrm{loc}}]\Big] \\ &= \mathbb{E}_D^{\mathrm{joint}}[\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}\mathbb{P}_D(\mathcal{F}_D \mid \mathscr{T}_{\mathrm{loc}})] . \end{aligned}$$

This is a tower identity for the existing joint law. It neither redefines D after localization nor requires independence between the event and the remaining randomness. □

Proposition A.21 (Integrated localized refinement failure)**.** *Under Assumptions 1, 2, and 3, Proposition A.20, and Lemma A.28, the exact estimator satisfies*

$$\boxed{\mathbb{P}_D(\mathcal{F}_D \cap \mathcal{E}_{\mathrm{loc}}) \le \frac{\delta}{4}.} \tag{13.7}$$

Proof. Proposition A.20 gives the almost-sure indicator inequality (13.1). Integrating it with respect to the marginal law of the full observable localization transcript and applying Lemma A.28 yields

$$\begin{aligned} \mathbb{P}_D(\mathcal{F}_D \cap \mathcal{E}_{\mathrm{loc}}) &= \mathbb{E}_D^{\mathrm{joint}}[\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}\mathbb{P}_D(\mathcal{F}_D \mid \mathscr{T}_{\mathrm{loc}})] \\ &\le \mathbb{E}_D^{\mathrm{joint}}\left[\frac{\delta}{4}\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}\right] \\ &= \frac{\delta}{4}\mathbb{P}_D(\mathcal{E}_{\mathrm{loc}}) \\ &\le \frac{\delta}{4}. \end{aligned} \tag{13.8}$$

The outer expectation accounts for every localization sample and seed; the conditional probability inside it already accounts for every refinement sample and seed. Thus (13.8) is under the same joint law as the goal. No extra independence step is inserted after the conditional certificate. □

Proposition A.22 (Unconditional uniform PAC conversion)**.** *Under Assumptions 1, 2, and 3, Lemma A.2, and Proposition A.21, for every* $D \in \mathcal{D}(k, \lambda, \sigma)$*,*

$$\boxed{\mathbb{P}_D\{|\widehat{\mu} - \mu(D)| > \epsilon\} \leq \frac{\delta}{4} + \frac{\delta}{4} = \frac{\delta}{2} \leq \delta.} \tag{13.9}$$

Consequently,

$$\sup_{D \in \mathcal{D}(k,\lambda,\sigma)} \mathbb{P}_D\{|\widehat{\mu} - \mu(D)| > \epsilon\} \leq \frac{\delta}{2} \leq \delta. \tag{13.10}$$

In the source trivial localization branch $2\lambda \leq 20\sigma$, the stronger unconditional bound $\mathbb{P}_D(\mathcal{F}_D) \leq \delta/4$ holds.

Proof. The events $\mathcal{F}_D \cap \mathcal{E}_{\text{loc}}$ and $\mathcal{F}_D \cap \mathcal{E}_{\text{loc}}^c$ are disjoint and their union is $\mathcal{F}_D$. Hence

$$\begin{aligned} \mathbb{P}_D(\mathcal{F}_D) &= \mathbb{P}_D(\mathcal{F}_D \cap \mathcal{E}_{\text{loc}}) + \mathbb{P}_D(\mathcal{F}_D \cap \mathcal{E}_{\text{loc}}^c) \\ &\leq \mathbb{P}_D(\mathcal{F}_D \cap \mathcal{E}_{\text{loc}}) + \mathbb{P}_D(\mathcal{E}_{\text{loc}}^c) \\ &\leq \frac{\delta}{4} + \frac{\delta}{4} = \frac{\delta}{2} \leq \delta. \end{aligned} \tag{13.11}$$

The first contribution is Proposition A.21; the second is Lemma A.2. The final inequality uses only $\delta > 0$.

The law D was arbitrary, and neither contribution nor its constant depends on which member of $\mathcal{D}(k, \lambda, \sigma)$ was fixed. Taking the supremum over that same unrestricted class proves (13.10). This supremum is taken only after the joint-probability inequality is proved for each fixed D; it is not interchanged with a conditional expectation.

If $2\lambda \leq 20\sigma$, Lemmas A.2 and A.3 give $c = 0$ and $|c - \mu(D)| \leq 10\sigma$ deterministically, so $\mathbb{P}_D(\mathcal{E}_{\text{loc}}^c) = 0$. Equation (13.11) then reduces exactly to $\mathbb{P}_D(\mathcal{F}_D) \leq \delta/4$. □

Completion of the unconditional confidence certificate. Lemma A.28 first verifies the required typing: the generated localization event belongs to the sigma-field of the full observable localization transcript, the failure event for the exact setting estimator is measurable under the joint law of all samples and seeds, and conditional probability means the conditional expectation of that failure indicator. It then proves the exact identity

$$\mathbb{P}_D(\mathcal{F}_D \cap \mathcal{E}_{\text{loc}}) = \mathbb{E}_D^{\text{joint}}[\mathbf{1}_{\mathcal{E}_{\text{loc}}} \mathbb{P}_D(\mathcal{F}_D \mid \mathscr{T}_{\text{loc}})] \, .$$

Proposition A.21 applies this identity to Proposition A.20. The indicator-valued bound integrates to a localized joint failure contribution at most $\delta/4$; no theorem-facing localization-success assumption remains.

Proposition A.22 then uses Lemma A.2 for the only other contribution, $\mathbb{P}_D(\mathcal{E}_{\text{loc}}^c) \leq \delta/4$, and the exact disjoint decomposition of the failure event. Therefore

$$\mathbb{P}_D\{|\widehat{\mu} - \mu(D)| > \epsilon\} \leq \delta/4 + \delta/4 = \delta/2 \leq \delta$$

for every D in the full moment class. Taking the supremum proves the unconditional PAC conclusion. The same estimator, common law, absolute error event, and full collection of sample and protocol randomness are used in both the conditional and unconditional statements. □

A.14 Rate and Protocol Closure

We collect the fixed quantities used in the rate calculation. The variance proxy and refinement allocation are

$$V_k = 3072\sigma^2 \mathbf{1}\{\mathcal{J}_\mathrm{f} \neq \varnothing\} + \mathbf{1}\{\mathcal{J}_\mathrm{c} \neq \varnothing\} \begin{cases} C^{\mathrm{c}}_{k,>}\sigma^2, & k > 2, \\ C^{\mathrm{c}}_2 \sigma^2 \log(\sigma/\epsilon), & k = 2, \\ C^{\mathrm{c}}_{k,<}\sigma^k H^{2-k}, & 1 < k < 2, \end{cases} \tag{14.1}$$

$$s = \left\lceil \frac{32V_k}{\epsilon^2} \right\rceil, \qquad q = 2\left\lceil 8\log\frac{4}{\delta}\right\rceil + 1, \qquad N_{\mathrm{ref}} = qs. \tag{14.2}$$

Set

$$t := \frac{\sigma}{\epsilon}, \qquad L := \log\frac{1}{\delta}, \qquad a := \log\frac{\lambda}{\sigma}, \tag{14.3}$$

and

$$g_k(t) := \begin{cases} t^2, & k > 2, \\ t^2 \log t, & k = 2, \\ t^{k/(k-1)}, & 1 < k < 2. \end{cases} \tag{14.4}$$

Then

$$r_k(\lambda, \sigma, \epsilon, \delta) = a + g_k(t)L. \tag{14.5}$$

Using the constants from the preceding variance bounds and the value of b_k fixed in Lemma A.5, define

$$B_k := \begin{cases} 3072 + C^{\mathrm{c}}_{k,>}, & k > 2, \\ 3072 + C^{\mathrm{c}}_2, & k = 2, \\ 3072 + C^{\mathrm{c}}_{k,<}(2b_k)^{2-k}, & 1 < k < 2, \end{cases} \tag{14.6}$$

$$Q_0 := 48 + \frac{3}{\log 2}, \qquad S_k := 32B_k + 1, \qquad R_k := Q_0 S_k, \tag{14.7}$$

and

$$L_0 := 30000 + \frac{1}{\log 2}, \qquad C_k := \max\{10000, L_0 + R_k\}. \tag{14.8}$$

All these quantities are positive and finite and depend at most on fixed k. The exact localization count and its uniform upper bound are

$$N_{\mathrm{loc}} = \begin{cases} 0, & 2\lambda \le 20\sigma, \\ \left\lceil 10000\left(\log\left\lceil\frac{2\lambda}{20\sigma}\right\rceil + \log\frac{4}{\delta}\right)\right\rceil, & 2\lambda > 20\sigma, \end{cases} \tag{14.9}$$

$$N_{\mathrm{loc}} \le 1 + 10000\log\frac{\lambda}{\sigma} + 30000\log\frac{1}{\delta}. \tag{14.10}$$

Finally, the unconditional confidence result established above is

$$\sup_{D\in\mathcal{D}(k,\lambda,\sigma)} \Pr_{D,\,\mathrm{protocol}}\{|\widehat{\mu} - \mu(D)| > \epsilon\} \le \frac{\delta}{2} \le \delta. \tag{14.11}$$

Proposition A.23 (Simultaneous Borel precommitment of the complete query bank)**.** *Under Assumptions 1 and 3, Proposition A.1, Lemma A.5, and Proposition A.5, every query in the exact two-block protocol is Borel. Moreover, the localization queries and all refinement queries can be and are jointly fixed, as one finite family, before the first response bit is observed.*

Proof. In the localization branch $2\lambda \le 20\sigma$, Proposition A.1 sets $I_{\text{loc}} = \varnothing$, so the localization Borel assertion is vacuous. In the nontrivial branch, the same proposition constructs each $\mathcal{B}_r$ as a finite union of clipped half-open or closed intervals. Thus every $\mathcal{B}_r$ is Borel. The codebook, clipped cells, enumeration, and least-index decoder convention depend only on known $(\lambda, \sigma, \delta)$, not on a sample or response. The instantiation takes R_{loc} degenerate, so the complete localization query family is deterministic and precommitted.

For refinement index i, condition only on its presampled seed $(L_i, A_i, B_i, U_i) = (j, a, b, u)$. Proposition A.5 proves that

$$\mathcal{A}_i = \left\{ x : \frac{F_{j,a,b}(x)}{h_j} \ge u \right\} \tag{14.12}$$

is Borel. More strongly, its membership indicator is jointly Borel in (x, j, a, b, u). The finite scale family and exact probabilities p_j are deterministic functions of known (k, σ, ϵ) and k-only constants. Assumption 3 requires every tuple (L_i, A_i, B_i, U_i), for every future refinement index, to be drawn before any response. Hence every realized set (14.12) is fixed at that same pre-message time.

There are exactly $N_{\text{loc}} + N_{\text{ref}} < \infty$ indices. First fix the deterministic localization bank and jointly draw the finite vector of all refinement seeds, then reveal no response until this is complete. This fixes the entire two-block query family simultaneously. No bank member contains the eventual decoded c, a previous bit, a block mean, or a stopping event. Therefore the complete query bank is Borel and fully precommitted. $\square$

Proposition A.24 (Exact one-bit, decoder-only, fixed-horizon protocol legality)**.** *Under Assumptions 1, 2, and 3, Proposition A.23, Lemma A.2, Lemma A.5, Lemma A.8, and Proposition A.20, the exact protocol has deterministic non-stopping horizon*

$$n = N_{\text{loc}} + N_{\text{ref}} = N_{\text{loc}} + qs. \tag{14.13}$$

It consumes exactly n independent samples and transmits exactly one bit from each sample. The localization output c, every selected shift or selected digit path, the centering threshold, and every importance weight are decoder-only. The dither baseline is a locally computed public-randomness quantity, not a second transmitted bit.

Proof. The two index sets are disjoint and have the deterministic sizes in (14.13). By Assumption 3, their samples are independent and share the same fixed law D.

For each $r \in I_{\text{loc}}$, the encoder observes only X_r and sends

$$Y_r = \mathbf{1}\{X_r \in \mathcal{B}_r\} \in \{0, 1\}. \tag{14.14}$$

There is one such message and no other message from X_r. In the zero-query branch, $I_{\text{loc}} = \varnothing$, so no nonexistent sample or bit is charged.

For each $i \in I_{\text{ref}}$, the encoder observes only X_i and the already public seed for its precommitted set, and sends

$$Y_i = \mathbf{1}\{X_i \in \mathcal{A}_i\} \in \{0, 1\}. \tag{14.15}$$

Again there is exactly one response. The decoder later forms

$$\beta_i(c) := \mathbf{1}\left\{ \frac{F_{L_i, A_i, B_i}(c)}{h_{L_i}} \ge U_i \right\}. \tag{14.16}$$

This is not another sample query. It is a deterministic Borel computation from the decoded scalar c, the known scales, and the same public seed used to define $\mathcal{A}_i$. Thus the exact pseudo-observation contains $Y_i - \beta_i(c)$, one transmitted bit minus one decoder-computed baseline. Neither U_i, the pair-match indicator, nor $p_{L_i}^{-1}$ is a response bit.

Lemma A.2 makes c a Borel function of the localization transcript. Lemma A.8 makes every $a_j(c)$ Borel. Consequently the selected pair, $Q_j^c, D_j^c, T_c, R_0^c, R_H^c$, pair-match test, centering in (14.16), reweighting, block averages, and odd median are all decoder or proof-analysis operations. The refinement query (14.12) uses the presampled pair (A_i, B_i), not the later selected pair $(a_{L_i}(c), a_{L_i+1}(c))$. Changing c after the bits arrive can change only which already observed bank entry is retained and how it is centered; it cannot change an encoder query.

The branch in (14.9) is determined from known (λ, σ). Lemma A.5 and Proposition A.2 determine J, H, the groups, and all p_j from known parameters. Lemma A.27 and Propositions A.18 and A.19 determine $V_k, s, q, N_{\mathrm{ref}}$ from known parameters and fixed k-only constants; none depends on a realized transcript or bit. The blocks are fixed before messages. The protocol therefore always uses the predetermined number (14.13), does not stop on localization success or a refinement event, and makes no fixed-to-random horizon conversion. This proves exact one-bit-per-sample, decoder-only center use, full nonadaptivity, and fixed-horizon legality. □

Lemma A.29 (Three-regime variance specialization). *Under Assumption 1, Lemma A.5, Proposition A.14, Proposition A.16, and Lemma A.27, the exact certificate V_k in (14.1) satisfies*

$$\boxed{\frac{V_k}{\epsilon^2} \le B_k g_k(t)} \tag{14.17}$$

in all three regimes. The inequality remains valid if either auxiliary group is empty, while under theorem design both groups are nonempty. At $k = 2$, (14.17) contains exactly one $\log(\sigma/\epsilon)$. It remains valid at $\epsilon = c_k\sigma = e^{-1}\sigma$.

Proof. Lemma A.5 gives

$$t = \frac{\sigma}{\epsilon} \ge e, \qquad H < 2H_* = 2b_k\sigma t^{1/(k-1)}. \tag{14.18}$$

The group indicators in (14.1) are at most one. If a group is absent, its term is exactly zero, so replacing its indicator by one can only enlarge the upper bound. Under theorem parameters both indicators equal one, so no actual theorem contribution is lost.

If $k > 2$, division of (14.1) by ϵ^2 gives

$$\frac{V_k}{\epsilon^2} \le (3072 + C^{\mathrm{c}}_{k,>})\frac{\sigma^2}{\epsilon^2} = B_k t^2 = B_k g_k(t). \tag{14.19}$$

If $k = 2$, then $\log t \ge 1$. The logarithm-free fine term is therefore absorbed by one, and only one, copy of the critical logarithm:

$$\begin{aligned} \frac{V_2}{\epsilon^2} &\le 3072t^2 + C_2^{\mathrm{c}} t^2 \log t \\ &\le (3072 + C_2^{\mathrm{c}})t^2 \log t = B_2 g_2(t). \end{aligned} \tag{14.20}$$

No other step in this calculation counts coarse levels. Proposition A.16 already proved that its coarse term has exactly one logarithm, and (14.20) only dominates the fine term by that same factor.

Finally let $1 < k < 2$. The top-scale inequality (14.18) gives

$$\begin{aligned}\frac{\sigma^k H^{2-k}}{\epsilon^2} &< (2b_k)^{2-k}\frac{\sigma^k\sigma^{2-k}}{\epsilon^2}t^{(2-k)/(k-1)}\\ &= (2b_k)^{2-k}t^{2+(2-k)/(k-1)}\\ &= (2b_k)^{2-k}t^{k/(k-1)},\end{aligned} \tag{14.21}$$

where

$$2+\frac{2-k}{k-1}=\frac{k}{k-1}. \tag{14.22}$$

Moreover

$$\frac{k}{k-1}-2=\frac{2-k}{k-1}>0,$$

and $t>1$, so $t^2 \le t^{k/(k-1)}$. Hence

$$\frac{V_k}{\epsilon^2} \le \left(3072 + C^{\mathrm{c}}_{k,<}(2b_k)^{2-k}\right)\, t^{k/(k-1)} = B_k g_k(t). \tag{14.23}$$

All coefficients are positive and finite for fixed k. Since $J \ge 1$, at least one group is nonempty; its coefficient and scale factor are positive, so $V_k > 0$. At the largest allowed error $t = e$, every power remains positive and the critical logarithm equals one. This proves every regime and boundary clause. □

Lemma A.30 (Refinement-count bound)**.** *Under Assumption 1, Lemma A.29, and the exact allocation in Lemma A.27, Proposition A.18, Proposition A.19, and Proposition A.20,*

$$\boxed{N_{\mathrm{ref}} = qs \le R_k g_k(t) L.} \tag{14.24}$$

This explicitly absorbs both ceilings, every additive one in the upper bounds for q and s, and the replacement of $\log(4/\delta)$ by $\log(1/\delta)$. The bound remains nondegenerate as $\delta \uparrow 1/2$.

Proof. Because $0<\delta<1/2$,

$$L = \log\frac{1}{\delta} > \log 2 > 0. \tag{14.25}$$

Also

$$\log\frac{4}{\delta} = L + \log 4 \le L + 2L = 3L, \tag{14.26}$$

because $\log 4 = 2\log 2 \le 2L$. Applying $\lceil x\rceil \le x+1$ to the exact odd block count gives

$$\begin{aligned} q &= 2\left\lceil 8\log\frac{4}{\delta}\right\rceil + 1\\ &\le 16\log\frac{4}{\delta} + 3\\ &\le 48L + 3\\ &\le \left(48 + \frac{3}{\log 2}\right) L = Q_0 L.\end{aligned} \tag{14.27}$$

The last line is valid throughout the allowed confidence domain because $L \ge \log 2$. Thus the $+3$, including the outer $+1$ making q odd and the doubled ceiling error, is charged explicitly to the positive target confidence logarithm.

For the exact block size, Lemma A.29 and the same ceiling inequality give

$$\begin{aligned} s &= \left\lceil \frac{32V_k}{\epsilon^2} \right\rceil \\ &\le \frac{32V_k}{\epsilon^2} + 1 \\ &\le 32B_k g_k(t) + 1. \end{aligned} \tag{14.28}$$

In all regimes $t \ge e$ implies $g_k(t) \ge 1$: this is immediate for t^2, for $t^2 \log t$ because $\log t \ge 1$, and for the positive power $t^{k/(k-1)}$. Therefore

$$s \le (32B_k + 1)g_k(t) = S_k g_k(t). \tag{14.29}$$

Multiplying (14.27) and (14.29), both nonnegative, proves (14.24). At $k = 2$, $g_2(t) = t^2 \log t$, so (14.24) has exactly one accuracy logarithm and one confidence logarithm, as in the public rate. As $\delta \uparrow 1/2$, $L \downarrow \log 2$, so the target factor remains positive and every ceiling remains absorbed by the displayed constants. □

Proposition A.25 (Rate Specialization Bridge for the fixed nonadaptive protocol)**.** *Under Assumptions 1, 2, and 3, Lemma A.3, Proposition A.22, Proposition A.24, and Lemma A.30, the exact deterministic horizon satisfies*

$$\boxed{n = N_{\text{loc}} + N_{\text{ref}} \le C_k r_k(\lambda, \sigma, \epsilon, \delta).} \tag{14.30}$$

The same exact estimator, without a probability, norm, horizon, or population scope change, satisfies

$$\boxed{\sup_{D \in \mathcal{D}(k,\lambda,\sigma)} \Pr_{D,\,\text{protocol}} \{|\widehat{\mu} - \mu(D)| > \epsilon\} \le \frac{\delta}{2} \le \delta.} \tag{14.31}$$

All hidden constants in (14.30) depend only on fixed k.

Proof. The auxiliary choices are exactly the ones: $c_k = e^{-1}$, b_k, H, V_k, and exact s, q in (14.2). Lemma A.29 verifies every scale condition needed to turn V_k/ϵ^2 into the public multiplier, and Lemma A.30 proves the complete refinement simplification (14.24).

It remains to absorb the exact localization cost. Lemma A.3 gives (14.9)–(14.10). Since $a = \log(\lambda/\sigma) \ge 0$, $g_k(t) \ge 1$, and $L \ge \log 2$,

$$1 \le \frac{1}{\log 2} g_k(t) L, \qquad L \le g_k(t) L. \tag{14.32}$$

Consequently every term, including the localization ceiling $+1$, obeys

$$\begin{aligned} N_{\text{loc}} &\le 1 + 10000a + 30000L \\ &\le 10000a + \left(30000 + \frac{1}{\log 2}\right) g_k(t) L \\ &= 10000a + L_0 g_k(t) L. \end{aligned} \tag{14.33}$$

Combining (14.24) and (14.33) gives

$$\begin{aligned} n &\le 10000a + (L_0 + R_k) g_k(t) L \\ &\le \max\{10000, L_0 + R_k\}\big(a + g_k(t)L\big) \\ &= C_k r_k(\lambda, \sigma, \epsilon, \delta), \end{aligned} \tag{14.34}$$

because both public-rate summands are nonnegative. This proves (14.30) with the explicit k-only constant in (14.8).

At $\lambda = \sigma$, $a = 0$ and, more strongly, $2\lambda \le 20\sigma$, so Lemma A.3 gives $N_{\mathrm{loc}} = 0$ exactly. Thus (14.34) does not rely on a positive localization logarithm: all remaining ceiling and confidence costs are charged to the positive refinement term $g_k(t)L$. Throughout the larger zero-query branch $2\lambda \le 20\sigma$, the equality $N_{\mathrm{loc}} = 0$ is preserved. At $\epsilon = c_k\sigma$, $t = e$, so $g_k(t) \ge 1$, and at $k = 2$, $\log t = 1$. Equation (14.25) handles δ arbitrarily close to $1/2$.

Proposition A.22 already proves (14.31) for the exact estimator under the exact joint law. This bridge only attaches the deterministic count (14.30); it does not rerun a union bound, enlarge $\delta/2$, change the norm, condition on localization, replace the full class by a subclass, or turn the horizon into a stopping time. In the zero-query localization branch it also retains the sharper failure bound $\delta/4$. Therefore the probability conversion and rate statement are simultaneously valid in the required modes. $\square$

Proposition A.26 (Exact baseline invariance of the final certificate)**.** *Under Assumptions 1, 2, and 3, Lemma A.3, Proposition A.2, Lemma A.8, Proposition A.6, Proposition A.14, Proposition A.16, and Lemma A.27, the final rate/protocol certificate preserves all of the following exact baseline clauses:*

1. If $2\lambda \le 20\sigma$, localization uses exactly zero samples, returns $c = 0$, and introduces no query, bit, ceiling charge, or stopping decision. 2. An absent fine or coarse group contributes exactly zero to every sum and variance certificate; no absent normalizer or inverse probability is evaluated. Under theorem parameters both groups are nonempty and receive their mass $1/2$. 3. For every real c, at $x = c$, every centered selected digit, $T_c(c)$, $R_0^c(c)$, $R_H^c(c)$, and the centered dither bracket is exactly zero. Hence if a refinement sample equals c, then $Z_i(c) = 0$ pathwise for every level, offset, and dither realization. 4. If $D = \delta_\mu$ and the generated center equals μ, every block mean and the odd median are zero, and $\widehat{\mu} = \mu$ exactly. No positive error remainder is inserted by the rate absorption.

Proof. Clause 1 is the exact zero-query branch in Lemma A.3; (14.33) is only an upper bound and does not replace that equality by one fictitious localization sample.

For Clause 2, Proposition A.2 defines weights and probabilities only on nonempty groups. Its empty-sum extension is zero, and Propositions A.14 and A.16, together with Lemma A.27, preserve that zero with explicit group indicators. The proof of (14.17) replaces indicators by one only to obtain an upper bound; it does not modify the implemented law or exact absent-group value. Under theorem design, endpoint witnesses put one sampled level in each group, so $m = 2$ and each group receives mass $1/2$.

For Clause 3, the setting definitions give, without an inequality,

$$D_j^c(c) - D_j^c(c) = 0, \qquad T_c(c) = \sum_{j=0}^{J-1} 0 = 0, \tag{14.35}$$

$$R_0^c(c) = (c - Q_0^c(c)) - (c - Q_0^c(c)) = 0, \qquad R_H^c(c) = Q_J^c(c) - Q_J^c(c) = 0. \tag{14.36}$$

Proposition A.6 gives the stronger pointwise dither identity: when $x = c$, the encoder threshold indicator and decoder baseline indicator coincide for every U. Directly in the actual pseudo-observation, if $X_i = c$, then (14.15) equals (14.16), so

$$Y_i - \beta_i(c) = 0 \quad \text{and hence} \quad Z_i(c) = 0 \tag{14.37}$$

for every value of the match indicator and importance weight. This also reconfirms that the baseline is a decoder computation rather than a second message.

For Clause 4, fix the point-mass law $D = \delta_\mu$ and suppose the generated center equals μ. Then every refinement sample equals $c = \mu$, so (14.37) holds at every refinement index. Every fixed block average and its odd median are zero, and the exact decoder returns $c+0 = \mu$. Lemma A.27 preserves the same zero-variance conclusion. None of (14.17), (14.24), or (14.34) changes the estimator, adds noise, or substitutes an $O(h_j)$ residual. Thus all four baseline clauses pass unchanged through the final certificate. □

Completion of the rate and protocol certificate. Proposition A.23 combines the localization and refinement interfaces and proves that the entire finite query bank is Borel and simultaneously fixed before any message. Proposition A.24 verifies the communication path: each independent sample produces exactly one membership bit; the public dither baseline is locally computed rather than transmitted; c, every stable selector, centered digit path, reweighting, block average, and median are decoder-only; and the exact horizon is deterministic and non-stopping.

Lemma A.29 uses the fine and coarse certificates and scale relation to prove $V_k/\epsilon^2 \le B_k g_k(t)$. It treats $k = 2$ directly and retains exactly one $\log(\sigma/\epsilon)$. Lemma A.30 starts from the exact ceilings in s, q and proves $N_{\mathrm{ref}} \le R_k g_k(t) L$, including every $+1$, the odd-median outer $+1$, the confidence substitution, and the boundary $\delta \uparrow 1/2$.

Proposition A.25, the named Rate Specialization Bridge, adds the exact localization count. It charges the localization ceiling and confidence term to the positive refinement budget when $\log(\lambda/\sigma) = 0$, preserves the exact zero-query source branch, and obtains

$$n \le C_k \left[\log\frac{\lambda}{\sigma} + \begin{cases} \dfrac{\sigma^2}{\epsilon^2}\log\dfrac{1}{\delta}, & k > 2, \\ \dfrac{\sigma^2}{\epsilon^2}\log\dfrac{\sigma}{\epsilon}\log\dfrac{1}{\delta}, & k = 2, \\ \left(\dfrac{\sigma}{\epsilon}\right)^{k/(k-1)}\log\dfrac{1}{\delta}, & 1 < k < 2 \end{cases} \right] = C_k r_k(\lambda,\sigma,\epsilon,\delta). \tag{14.38}$$

The same proposition consumes Proposition A.22 without modification and therefore carries the exact unconditional, absolute-value, fixed-horizon, full-population statement (14.31). Proposition A.26 confirms that the upper-bound algebra does not weaken the exact zero-query, empty-group, zero-displacement, dither, or fixed-point-mass-law specialization with generated center equal to its mean.

These six named results prove every stated conclusion: Borel simultaneous precommitment, exactly one bit per independent sample, decoder-only use of c, deterministic fixed horizon, exact three-regime sample complexity with only k-dependent constants, the unchanged unconditional PAC event, and all inherited exact baselines. No new theorem- facing assumption, encoder query, interaction, protocol mechanism, cited result, probability conversion, or conclusion is introduced. □

A.15 Proof of the Main Theorem

Proof of Theorem 3.1. Lemma A.5 verifies that the displayed k-only choices are finite and legal, that $J \ge 1$, and that the refinement bank's scale family has the required endpoint ordering. The exact variance constants are established in Propositions A.14 and A.16; Lemma A.27 combines them into the displayed V_k, and Propositions A.18 and A.19 justify the stated s and q.

Proposition A.23 proves that the entire finite query bank is Borel and fixed before communication. Proposition A.24 then gives exactly one transmitted bit per independent sample, decoder-only use of c, and the deterministic horizon $n = N_{\mathrm{loc}} + qs$. Proposition A.22 proves the displayed unconditional uniform failure bound for the same estimator and joint law. Finally, Proposition A.26

preserves the exact zero-query, zero-displacement, and point-mass conclusions. These statements are precisely the conclusions of the theorem. □

Proof of Corollary 3.1. Apply Theorem 3.1 with its displayed choices. The Rate Specialization Bridge, Proposition A.25, proves for the same fixed-horizon protocol that

$$n \le C_k r_k(\lambda, \sigma, \epsilon, \delta)$$

with C_k depending only on k, while retaining the unconditional absolute-error guarantee of Proposition A.22. Proposition A.24 retains simultaneous precommitment, one bit per independent sample, and decoder-only use of the localization center. This proves every assertion of the corollary. □